CHAPTER 2

The Shell Model

2.1 Introduction and General Considerations

In the last chapter we considered the bulk properties of the nucleus, that is, we discussed (static or dynamic) properties for which at least a good fraction of all the nucleons in a nucleus participate. In this chapter we are going to talk about a completely different aspect of the nucleus. Indeed, many nuclear properties seem to be describable in terms of the idea that the nucleons in a nucleus are to be considered as independent particles moving on almost unperturbed single particle orbits. The reasons for this, as we stated at the beginning of the first chapter, is the fact that, mostly due to the action of the Pauli and uncertainty principles, the nucleus is not a very dense system. It is now quite well established that the nucleon–nucleon force has an almost infinitely repulsive core (see [Vi 78]) at a radius c of about $c=0.4$ fm. Therefore, the ratio of the closest packed volume V_c to the actual volume V of a nucleus is [BM 69, Sec. 2.5]

$$\frac{V_c}{V}=\left(\frac{c}{2r_0}\right)^3\simeq\frac{1}{100}\,.$$

Thus the known "strong" character of the nucleon–nucleon forces is considerably reduced by the fact that the nucleons are, on the average, quite far apart, and therefore "feel" only the tail of the attractive part of the nuclear force. In other words, the violent interactions due to the singular force occur only quite seldom and the system can be described, at least in a first approximation, in terms of independent particle motion. The

fact that despite these considerations the nucleus develops a very well-defined surface, contrary to a gas, is due to a very subtle interplay of the nuclear forces and the Pauli principle [BP 77a]. For a further, more elaborate discussion of these considerations, see also [BM 69, Sec. 2.5]. The mean free path of the nucleons in a nucleus, as can be estimated from scattering experiments, seems to be at least of the order of the dimension of the nucleus [BM 69, Sec. 2.1; KK 68] and is mentioned as the first piece of experimental evidence for the unperturbed particle motion in a nucleus.

The idea of independent particles accepted, it is quite natural to envisage that this single particle motion is governed by some average potential created by all the nucleons in the nucleus. Of course, the motion of the nucleons will be considerably different in the interior of the nucleus, where it is more or less force free, from the one at the surface where the Pauli principle ceases to act and the particles feel a force confining them to the interior of the nucleus.

In this chapter we will briefly describe further experimental evidence for, and the phenomenological description and consequences of, such an average potential.

2.2 Experimental Evidence for Shell Effects

If the validity of an average potential in which the nucleons can move independently can be assumed, this immediately has some obvious consequences similar to those with which we are familiar from atomic physics.

The occurrence of the so-called magic numbers 2, 8, 20, 28, 50, 82, and 126 has, from the experimental point of view, been one of the strongest motivations for the formulation of a nuclear shell model. At these proton or neutron numbers, effects analogous to shell closure of electron shells in atoms are observed. Here we will mention just a few of them.

The single-particle separation energy is defined as the energy required to remove the least bound particle from the nucleus. In Fig. 2.1 the observed separation energies for protons and neutrons are shown.

For most nuclei the separation energy is about 8 MeV, although there are quite prominent exceptions at the magic numbers. The separation energy is largest for doubly magic nuclei. Similar exceptions for the separation energy are found for the electrons in noble gases.

The magic numbers can be seen in Fig. 2.2, which shows that the magic and doubly magic nuclei are exceptionally strongly bound. Strong binding means that the nucleus is very stable against excitations, and in Fig. 1.7 we have already shown the variation of the first 2^+ state in nuclei as a function of nucleon number. We can see especially pronounced shell effects at the magic numbers, the excitation energy rising sharply in the neighborhood of shell closures. Other collective excitations show the same variation. These examples should be sufficient as a demonstration of the occurrence of magic numbers in nuclei and of the shells corresponding to

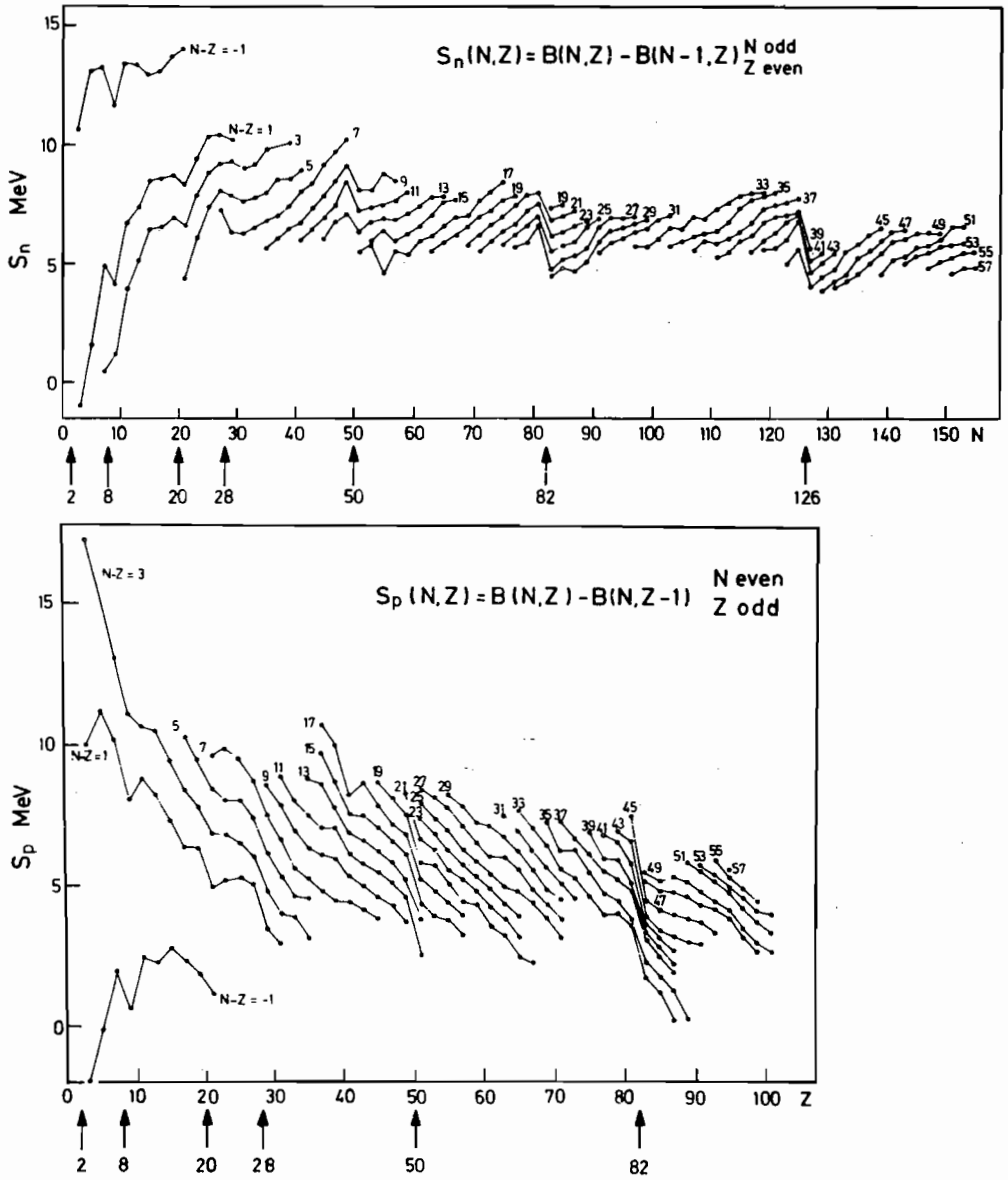


Figure 2.1. Neutron and proton separation energies as a function of neutron and proton number, respectively. (From [Ir 72].)

an average potential analogous to the ones observed in atoms. (More details may be found in [MJ 55].)

2.3 The Average Potential of the Nucleus

As mentioned above, the exceptional role of the magic numbers reveals a strong analogy to the situation of the electrons in an atom. There the strong central Coulomb potential of the nucleus imposes sphericity. As a consequence, there exist groups of degenerate levels with quite large energy differences in between the electron shells.

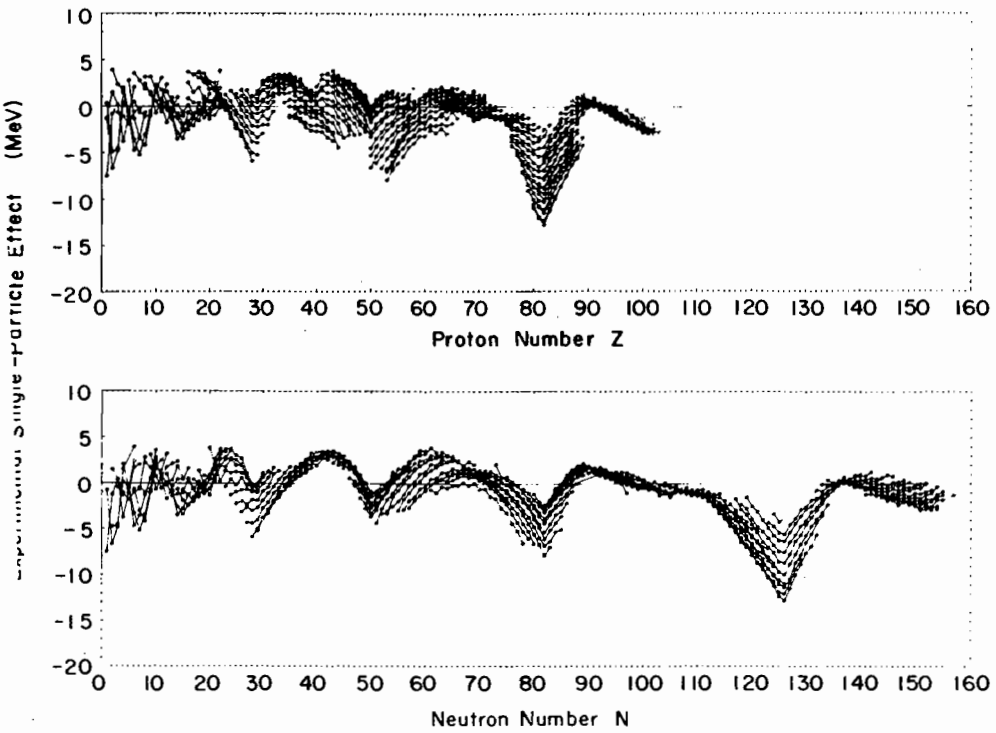


Figure 2.2. Deviations of nuclear masses from their mean values plotted as a function of neutron and proton number. (From [MS 66].)

For the nucleons of a nucleus, there exists a priori no such central field. As we have already discussed in the introduction, however, we can imagine such a potential as being built up by the action of all the nucleons. (Such an average potential also exists in the case of the electrons in an atom; there it makes an extra contribution and has to be added to the nuclear Coulomb potential. The whole is then well known as the Hartree or Hartree–Fock potential of the atom.) A model which describes the dynamics of the nucleons only with such an average potential treats the nucleons as completely independent of one another (the nucleon–nucleon interaction nevertheless comes into it in an indirect way, since it gives rise to the average potential in the first place). In the following, we will call such a model the shell model or independent particle model.

In Chapter 5 we will discuss, in great detail, how one can derive the form of this average field from a microscopic two-body force. In this chapter, we will assume that we have such a one-body potential which describes, to a good approximation, the effects of the mutual interaction between the nucleons, and investigate their consequences.

For further consideration, we have to discuss first the shape of the ad hoc introduced shell model potential. A nucleon close to the center of the nucleus will feel the nuclear forces uniformly, that is, there will be no net force

$$\left(\frac{\partial V(\mathbf{r})}{\partial r}\right)_{r=0} = 0. \tag{2.1}$$

The nuclear binding forces get stronger going from the surface ($r = R_0$) to the interior of the nucleus:

$$\left(\frac{\partial V}{\partial r}\right)_{r<R_0} > 0. \tag{2.2}$$

Because of the finite range of the nuclear forces, we have:

$$V(r) \simeq 0, \qquad r > R_0. \tag{2.3}$$

An analytic ansatz which represents these conditions quite well, and also yields quite reasonable density distributions, is the Fermi function or *Woods–Saxon potential* [WS 54] (Fig. 2.3):

$$V^{\text{W.S.}}(r) = -V_0\left[1+\exp\left(\frac{r-R_0}{a}\right)\right]^{-1} \tag{2.4}$$

with

$$R_0 = r_0 A^{1/3}; \qquad V_0 \simeq 50\ [\text{MeV}]; \qquad a \simeq 0.5\ [\text{fm}]; \qquad r_0 \simeq 1.2\ [\text{fm}].$$

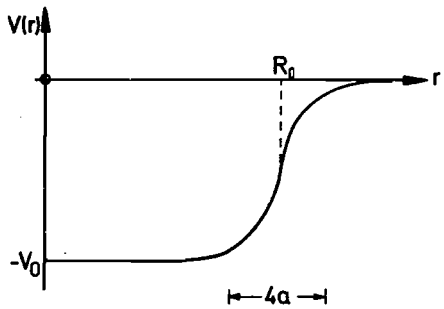


Figure 2.3. Shape of the Woods–Saxon potential.

(The Woods–Saxon potential actually has a finite but negligible slope at $r = 0$.) Since the eigenfunctions for this potential cannot be given in closed form, one often uses the following two approximations for qualitative considerations, and also for calculations:

(i) harmonic oscillator

$$V(r) = -V_0\left[1-\left(\frac{r}{R_0}\right)^2\right] = \frac{m}{2}\omega_0^2(r^2 - R_0^2) \tag{2.5}$$

(ii) square well

$$V(r) = \begin{cases} -V_0 & \text{for } r \leqslant R_0, \\ +\infty & \text{for } r > R_0. \end{cases} \tag{2.6}$$

Before we discuss the solutions of Eqs. (2.4), (2.5), and (2.6) in more detail, we should perhaps note that all three potentials are spherically symmetric. For the moment, we will restrict our considerations to this case; the discussion of deformed potentials will be taken up in Section 2.8. Furthermore, it should be pointed out that (2.5) and (2.6) represent somewhat unphysical potentials, since they are infinite. However as long as we are

only interested in bound single-particle states, this is not too serious a drawback, as only the exponential tails of the wave functions are affected. If one considers excitations in these potentials, however, one easily gets into regions in which the states in the realistic potential (2.4) would be in the continuum. The use of infinite potentials in such cases is then to be considered with extreme care.

After these preliminary remarks, we want to discuss the energy levels obtained from the solution of the eigenvalue problem

$$\left\{-\frac{\hbar^2}{2m}\Delta + V(\mathbf{r})\right\}\phi_i(\mathbf{r}) = \epsilon_i\phi_i(\mathbf{r}) \tag{2.7}$$

for the case of the potentials (2.5) and (2.6).

As is well known, the harmonic oscillator gives equidistant energy levels

$$\epsilon_N = \hbar\omega_0\left(N+\frac{3}{2}\right) - V_0 \tag{2.8}$$

with

$$N = 2(n-1)+l, \qquad \text{where } n = 1, 2, \ldots, \text{ and } l = 0, 1, 2, \ldots. \tag{2.9}$$

These levels are $D(N)$-fold degenerate:

$$D(N) = \frac{1}{2}(N+1)(N+2), \tag{2.10}$$

where N is the number of quanta in the oscillator, n is the radial quantum number, l is the angular momentum, and ω_0 is the oscillator frequency. The oscillator constant ω_0 is usually determined from the mean square radius of a sphere

$$\langle R^2\rangle = \frac{1}{A}\sum_{i=1}^{A}\langle r^2\rangle_i \simeq \frac{3}{5}(1.2A^{1/3})^2[\mathrm{fm}^2]. \tag{2.11}$$

For oscillator states we can calculate $\langle r^2\rangle_i$ and get

$$\frac{m}{2}\omega_0^2\langle r^2\rangle_i = \frac{\hbar\omega_0}{2}\left(N_i+\frac{3}{2}\right).$$

Together with Eq. (2.10), we find [Mo 57, p. 469]

$$\hbar\omega_0 \simeq \frac{5}{4}\left(\frac{3}{2}\right)^{1/3}\frac{\hbar^2}{mr_0^2}A^{-1/3} = 41\cdot A^{-1/3}[\mathrm{MeV}] \tag{2.12}$$

and for the oscillator length

$$b = \sqrt{\frac{\hbar}{m\omega_0}} = 1.010\cdot A^{1/6}\,[\mathrm{fm}]. \tag{2.13}$$

The levels with a definite N we call an oscillator shell. Because of Eq. (2.9), the oscillator shells only contain either even or odd l-values, that is one oscillator shell contains only states with the same parity. It also follows from (2.9) that levels with the same N and with different n and l are degenerate. This accidental degeneracy of the harmonic oscillator is removed in the square-well potential (Fig. 2.4). The true energies lie between the two limits given by the potentials of Eqs. (2.5) and (2.6).

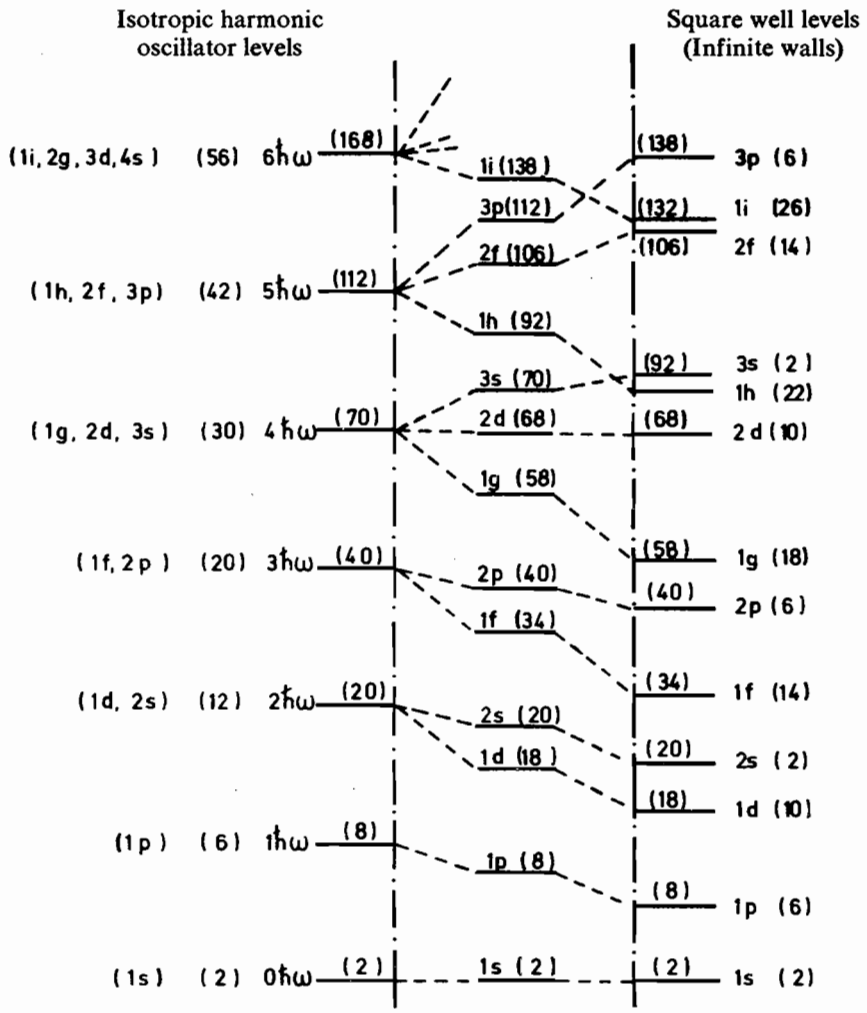


Figure 2.4. Level scheme of the isotropic harmonic oscillator (l.h.s.) and of the infinite square well (r.h.s.). (From[MJ 55]).

Filling up the level scheme with nucleons (by analogy with the periodic system of the atoms), we see that according to the Pauli principle, $D(N)$ protons and $D(N)$ neutrons can be put into each oscillator shell. This means that both potentials reproduce the magic numbers 2, 8, and 20. This model can therefore account for the unusual stability of ${}^{4}_{2}He_2$, ${}^{16}_{8}O_8$, and ${}^{40}_{20}Ca_{20}$. On the other hand, Fig. 2.4 contains no indications for the higher magic numbers. We will see in the next section how this deficiency of this simple model can be removed. Later, we will also discuss how the Coulomb interaction of the protons influences the average potential (Sec. 2.5).

2.4 Spin Orbit Coupling

Up to now, we have not taken into account the spin of the nucleons (apart from a factor of 2 in determining the magic numbers), that is, we considered the nuclear forces as spin independent. Treating the electrons in

atoms relativistically yields a spin dependent force in the form of a spin orbit coupling

$$f(r)\mathbf{ls}. \tag{2.14}$$

This gives a splitting of the otherwise degenerate levels with $j = l \pm \frac{1}{2}$. In the nucleus such a splitting has also been found experimentally. Scattering of protons or neutrons on α particles yields for the lowest resonance the (unbound) ground state of ^{5}Li or ^{5}He. According to Fig. 2.4 these lowest resonances should have the quantum numbers $l = 1$, $j = \frac{1}{2}, \frac{3}{2}$, as the 1s-shell in ^{4}He is closed. One observes resonances at 1.25 and 2.4 MeV for the scattering of neutrons and protons, respectively. At these energies the angular distributions show that the resonances are predominantly $j = \frac{3}{2}$, whereas the $j = \frac{1}{2}$ resonances lie a few MeV higher in energy.

It was a decisive idea (Haxel, Jensen, and Suess [HJS 49]; Goeppert–Mayer [Ma 49]) to incorporate a strong spin orbit term into the single-particle Hamilton operator of Eq. (2.7). It was only then that the success of the shell model was confirmed. Mathematically this yields a *jj*-coupling scheme, since $\mathbf{l}\cdot\mathbf{s}$ commutes with $\mathbf{s}^2, \mathbf{l}^2, \mathbf{j}^2, j_z$ but not with l_z and s_z. The levels are characterized in this coupling scheme by the quantum numbers n, l, j (e.g., $(2d\frac{5}{2}) \triangleq (n=2, l=2, j=2+\frac{1}{2})$), and a single particle wave function takes the following form*:

$$\phi(\mathbf{r})_{nsljm} = \langle \mathbf{r} | nsljm \rangle = \phi_{nl}(r) \sum_{m_l m_s} C^{\frac{1}{2}\ l\ j}_{m_s\ m_l\ m} Y_{lm}(\theta, \phi) | \tfrac{1}{2} m_s \rangle. \tag{2.15}$$

With the relations

$$\begin{aligned} 2\mathbf{ls}|sljm\rangle &= (\mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2)|sljm\rangle \\ &= \left[j(j+1) - l(l+1) - \tfrac{3}{4} \right] |sljm\rangle \end{aligned} \tag{2.16}$$

we are able to give the spin orbit splitting of the doubly degenerate levels $|slj = l \pm \frac{1}{2}\rangle$ for $f(r) = \text{const.}$:

$$\Delta E(l) \sim \left[l - (-l-1) \right] = 2l + 1. \tag{2.17}$$

An attractive spin orbit potential will assure the experimentally observed fact that the $l + \frac{1}{2}$ levels are energetically always below the $l - \frac{1}{2}$ levels. Equation (2.17) shows furthermore that the splitting increases with growing values of l.

Inclusion of the spin orbit interaction to the interpolated level scheme of Fig. 2.4 yields the modified level scheme of Fig. 2.5. The model now reproduces all magic numbers correctly. The sequence of the levels within the different shells depends on the choice of $f(r)$.

The value of $f(r)$ which one could derive by analogy with the theory of electrons in an atom using a Lorentz invariant treatment of the electromagnetic interactions of the nucleons (Thomas term: cf. [MJ 55, p. 60] and [EG 70, Chap. 8]) turns out to be about an order of magnitude too small.

* For calculations one has to pay particular attention to the coupling order of the angular momenta, since a different order introduces a phase which is the source of frequent errors.

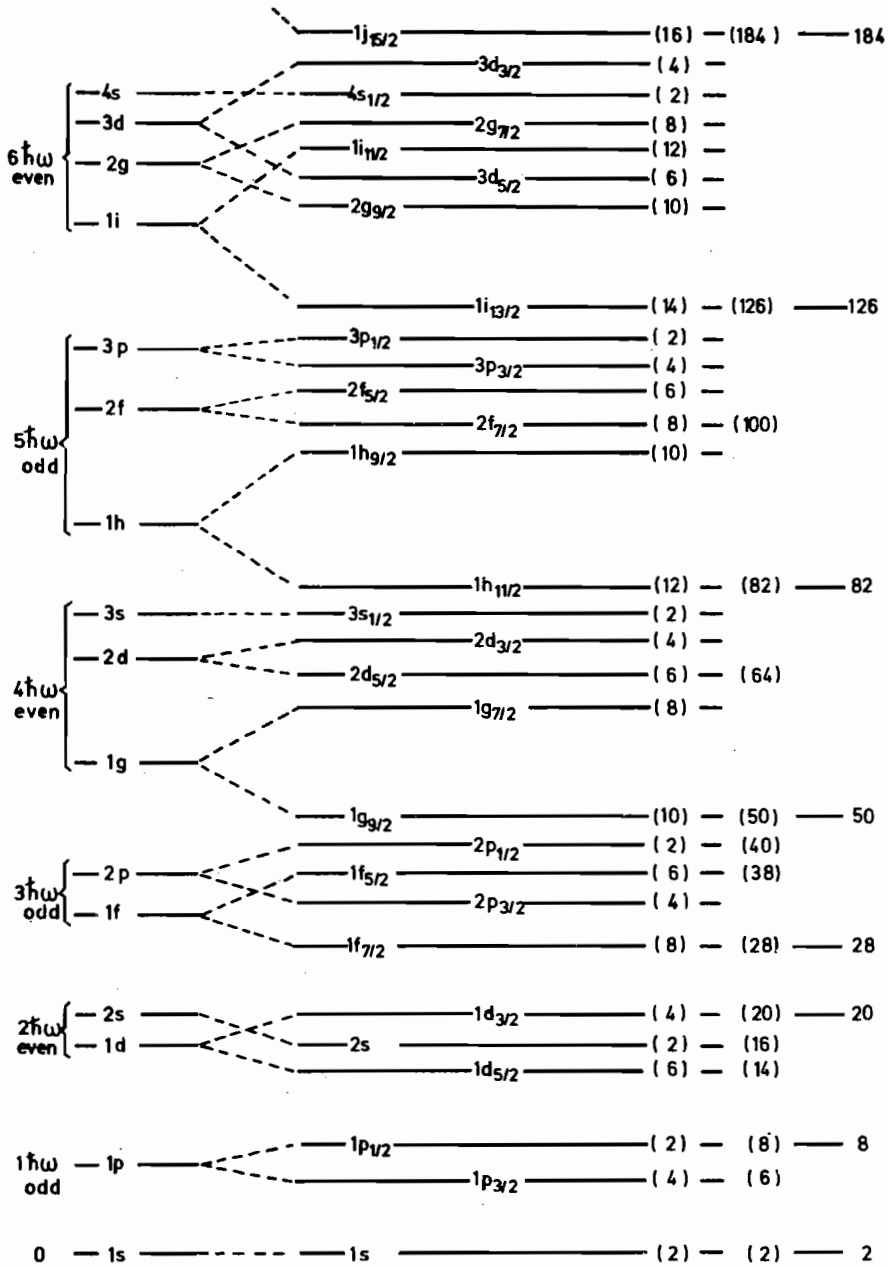


Figure 2.5. Schematic nuclear levels of the shell model with spin orbit term. (From [MJ 55].)

As we will discuss in Chapter 4, the nucleon–nucleon forces exhibit strong spin orbit parts, and it is thus not very surprising that the average single particle potential also has a spin orbit part with a strength compatible with the strong nuclear forces.*

* Recent investigations have shown that one obtains the proper spin orbit term in the single-particle potential by a relativistic Hartree–Fock treatment of one-boson-exchange-potentials [Br 78].

One can show [Ho 75] that $f(r)$ is peaked at the nuclear surface. By analogy with the electronic case, one quite often chooses $f(r)$ related to the spin independent part of the average potential in the following way—

$$f(r)=\lambda\frac{1}{r}\frac{dV}{dr}; \qquad \lambda\simeq -0.5\ [\mathrm{fm}^2] \tag{2.18}$$

—but also other surface-peaked radial dependences of $f(r)$ can be envisaged. It is interesting to note that the use of the Skyrme force (see Chap. 5) yields a spin orbit dependence for the average potential with $f(r)\sim(1/r)\cdot(d\rho/dr)$, where ρ is the single particle density. Since $V(r)$ roughly follows the form of ρ, this is consistent with Eq. (2.18).

2.5 The Shell Model Approach to the Many-Body Problem

The single-particle model takes into account the individual nucleons. It therefore provides a microscopic description of the nucleus. This is certainly only an approximation of the exact many-body problem. We will see, however, in the following, that the shell model can be used as a basis for more elaborate many-body theories, so before we talk about further details of the model, we want to discuss some general properites of the single particle model.

The microscopic theory of the nucleus is usually based on the following three properties.

(i) The nucleus is a quantum mechanical many-body system.
(ii) The velocities in the nucleus are small enough so that one can neglect relativistic effects $[(v/c)^2\sim 1/10]$.
(iii) The interaction between the nucleons has a two-body character.

A full microscopic theory of the nucleus would then be given by the solution of the many-body Schrödinger equation

$$H\Psi=\left\{\sum_{i=1}^{A}-\frac{\hbar^2}{2m}\Delta_i+\sum_{i<j}^{A}v(i,j)\right\}\Psi(1,\ldots,A)=E\Psi(1,\ldots,A), \tag{2.19}$$

where i represents all coordinates of the ith nucleon, for instance,

$$(i)=(\mathbf{r}_i, s_i, t_i), \tag{2.20}$$

where t_i will be $\frac{1}{2}$ for neutrons and $-\frac{1}{2}$ for protons. With the assumption of the nuclear shell model, the above equation reduces to the much simpler equation

$$H_0\Psi=\left\{\sum_{i=1}^{A}h_i\right\}\Psi=\sum_{i=1}^{A}\left\{-\frac{\hbar^2}{2m}\Delta_i+V(i)\right\}\Psi=E\Psi. \tag{2.21}$$

The solutions Ψ of Eq. (2.21) are anti-symmetrized products of single-particle functions, which are eigenfunctions to the single-particle Hamilto-

nian h_i:

$$h_i\phi_k(i)=\epsilon_k\phi_k(i). \tag{2.22}$$

The functions ϕ_k provide an orthogonal basis for an occupation number representation within the framework of second quantization (see Appendix C). To each level k corresponds a pair of creation and annihilation operators a_k^+, a_k which create or annihilate particles with wave function ϕ_k. Since nucleons are Fermions, each level can be occupied only once, and the operators a_k, a_k^+ obey Fermi commutation relations (C. 23).

The shell model Hamiltonian H_0 has the form

$$H_0=\sum\epsilon_k a_k^+ a_k.$$

Using the bare vacuum $|-\rangle$ its eigenfunctions can be represented as

$$|\Phi_{k_1\ldots k_A}\rangle=a_{k_1}^+\ldots a_{k_A}^+|-\rangle.$$

They are Slater determinants

$$\Phi_{k_1\ldots k_A}(1,\ldots,A)=\begin{vmatrix}\phi_{k_1}(1) & \cdots & \phi_{k_1}(A)\\ \vdots & & \vdots\\ \phi_{k_A}(1) & & \phi_{k_A}(A)\end{vmatrix} \tag{2.23}$$

with eigenvalues

$$E_{k_1\ldots k_A}=\epsilon_{k_1}+\cdots+\epsilon_{k_A}. \tag{2.24}$$

In the ground state the levels are filled successively according to their energy (see Fig. 2.6)

$$|\Phi_0\rangle=a_1^+\ldots a_A^+|-\rangle. \tag{2.25}$$

Thus we have for closed shells the following unique prescription for the construction of the A particle ground state as well as for the A particle excitation spectrum: Starting with the $(1s_{1/2})$ level, one has to occupy each level $|nsljm\rangle$ with just one particle until all A particles are used up. We thus obtain an A nucleon ground state where all different quantum states are occupied with just one particle up to the Fermi level (the highest occupied level); above the Fermi level all levels are unoccupied.

The independent particle picture of the nucleus is different from that in an atom in the sense that in a nucleus there are *two* different kinds of particles, the proton and the neutron, whereas in an atom there is only the

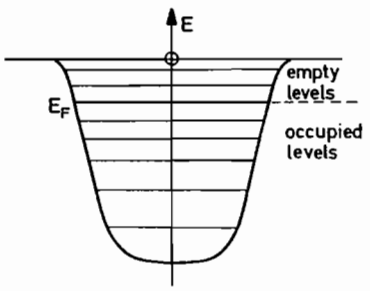


Figure 2.6. Shell model potential and Fermi level.

electron. Protons and neutrons feel different average potentials for two reasons:

(i) Protons also interact via the *Coulomb force*. One therefore usually adds the potential of a homogeneously charged sphere

$$V_C(\mathbf{r}) = \begin{cases} \dfrac{Ze^2}{R}\,\dfrac{1}{2}\left(3-\left(\dfrac{r}{R}\right)^2\right) & r \leqslant R, \\ \dfrac{Ze^2}{r} & r > R. \end{cases} \tag{2.26}$$

Sometimes (see Sec. 2.8), this feature is approximated by using different potential parameters for protons and neutrons.

(ii) The *symmetry energy* [see Eq. (1.4)] favors a configuration with an equal number of protons and neutrons. Because of the Coulomb repulsion for heavier nuclei, one has a neutron excess: If, in the nucleus, we replace a neutron by a proton, we gain symmetry energy and lose Coulomb energy. Since the Coulomb energy is already taken into account by Eq. (2.26), there must be an additional difference between the single-particle potential for protons and neutrons, which is caused by the symmetry energy. The *nuclear* part of the proton potential is therefore deeper (see Fig. 2.7, dashed line).

These two effects go in opposite directions, but they do not cancel. In the end, the Fermi surfaces for protons and neutrons must be equal, otherwise protons would turn into neutrons by β-decay or vice versa, whichever is energetically favored.

In $N \neq Z$ nuclei, energy levels with the same quantum numbers for protons and neutrons are therefore shifted with respect to one another by an amount Δ_ϵ resulting from a positive contribution Δ_C from the Coulomb force and a negative contribution $-\Delta_S$ from the symmetry energy

$$\epsilon_{nlj}^{(p)} - \epsilon_{nlj}^{(n)} = \Delta_\epsilon = \Delta_C - \Delta_S. \tag{2.27}$$

In heavy nuclei, this difference is such that the protons and neutrons at the Fermi surface belong to different major shells.

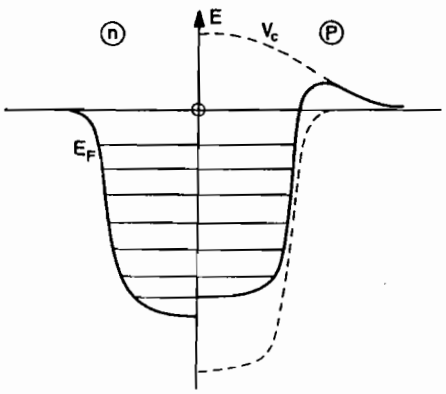


Figure 2.7. Comparison of the shell model potential for neutrons and protons in a nucleus with neutron excess.

Strong support of the independent particle idea comes from the experimental fact that magic numbers are the same for protons and neutrons (see Fig. 2.2; the magic number 126 exists only for neutrons, since for the heaviest nucleus known so far we only have $Z=103$). If correlations played a major role, then, for example, the neutron excess in heavier nuclei would eventually influence the proton magic numbers in these nuclei (the nuclear force is almost charge independent; see Chap. 4) to be different from the corresponding neutron numbers. However, as we have said, this is not the case. The subshells of a major shell have, in some cases, a different order.

In the shell model, the *excitations of the system* are given by analogy with the free Fermi gas by a transfer of nucleons from below the Fermi level to levels above it. In the case of only a single nucleon transfer, we talk of an $1p-1h$ state with excitation energy of $\sim\hbar\omega_0$. For $^{40}_{20}Ca_{20}$ such a state is, for example, given by

$$(2s\tfrac{1}{2})^{-1}(1f\tfrac{7}{2}).$$

The Fermi level coincides in this case with the $1d_{3/2}$ level (see Fig. 2.5).

If we use the indices i, j for the levels below the Fermi surface ($\epsilon_i \leqslant \epsilon_F$), and the indices m, n for the levels above the Fermi surface ($\epsilon_n > \epsilon_F$), the lowest excitations in the shell model are then *ph* excitations of the form

$$|\Phi_{mi}\rangle := a_m^+ a_i|\Phi_0\rangle = \pm a_m^+ a_1^+ \ldots a_{i-1}^+ a_{i+1}^+ \ldots a_A^+|-\rangle \qquad (2.28)$$

with excitation energy $\epsilon_{mi}=\epsilon_m-\epsilon_i$.

In fact one has observed such states in magic nulcei. They are, however, not the lowest states. As we have already seen in Chapter 1, there are low-lying collective states which cannot be explained in the independent particle model.

The Slater determinants (2.23) form a *complete set* of states for the A nulceon system [Lö 55]. Each state of the system is characterized by the distribution of the nucleons among the levels of the single particle potential, that is, by the "occupation numbers" of the levels. It is usual to classify all excited states by taking the ground state as a reference state. The nucleons that are missing in the ground state are denoted by holes, those above the Fermi levels by particles. A typical multiparticle–multihole configuration is shown in Fig. 2.8

Starting from a magic nucleus with the mass number A, we can add a particle and obtain a nucleus with the mass number $A+1$. If we put the

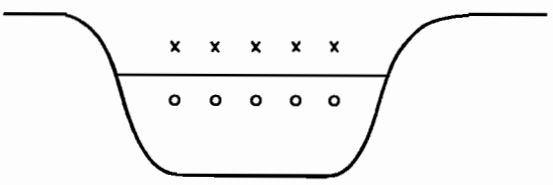


Figure 2.8. Schematic representation of a typical five-particle (crosses), five-hole (open circles) state.

particle into the level m, the wavefunction is

$$|\Phi_m\rangle = a_m^+ |\Phi_0\rangle \tag{2.29}$$

and we get the energy difference

$$\epsilon_m = E_m(A+1) - E_0(A). \tag{2.30}$$

In this way one is able to measure the single-particle energies (see Sec. 2.7). These are the simplest states in $A+1$ nuclei. More complicated states have a $2p-1h$ structure, and so on. In complete analogy, there are $1h$, $1p-2h$, etc., states in $A-1$ nuclei.

It often turns out to be very convenient to define quasiparticles by the operators

$$\begin{aligned} &\alpha_m^+ = a_m^+, && \alpha_m = a_m, && \text{for } \epsilon_m > \epsilon_F; \\ &\alpha_i^+ = a_i, && \alpha_i = a_i^+, && \text{for } \epsilon_i \leqslant \epsilon_F. \end{aligned} \tag{2.31}$$

These quasiparticles are again fermions. They are particles for states above, and holes for states below, the Fermi surface, so that we have

$$\alpha_k |\Phi_0\rangle = 0, \tag{2.32}$$

that is, the ground state of the magic nucleus is a "vacuum" with respect to these quasi-particles; *ph* states are two-quasi-particle states, and so on. The multi-quasi-particle states

$$|\Phi_{k_1 \ldots k_N}\rangle = \alpha_{k_1}^+ \ldots \alpha_{k_N}^+ |\Phi_0\rangle \tag{2.33}$$

form a complete orthogonal set in the many-body Hilbert space.

This basis is often used for further investigation of the many-body Hamiltonian H (2.19). In the shell model, one decomposes H,

$$H = T + \sum_{i<j} v(i,j) = H_0 + V_R, \tag{2.34}$$

with the residual interaction

$$V_R = \sum_{i<j} v(i,j) - \sum_i V(i) \tag{2.35}$$

in such a way that V_R is as small as possible and can be neglected. More elaborate theories investigate V_R in the basis in which H_0 is diagonal, the shell model basis (see Chap. 8).

The exact ground state wave function of a magic nucleus has the form

$$|\Psi_0\rangle = C_0 |\Phi_0\rangle + \sum_{mi} C_{mi} \alpha_m^+ \alpha_i^+ |\Phi_0\rangle + \tfrac{1}{4} \sum_{\substack{mi \\ m'i'}} C_{mim'i'} \alpha_m^+ \alpha_{m'}^+ \alpha_i^+ \alpha_{i'}^+ |\Phi_0\rangle + \cdots. \tag{2.36}$$

If the shell model is a good approximation to the nucleus, the coefficients C_{mi}, $C_{mim'i'}$, etc. should be small (see Fig. 10.3).

At this point we would like to again stress the fact that we have always been talking about a *spherical shell model* potential. Since, as we shall see, spherical nuclei exist only in the neighborhood of magic nuclei, by the same token this means that we have restricted our discussion to such nuclei. As this spherical average potential is created by the nucleons themselves, it may depend (though not abruptly) on the nucleon number A in quite a subtle way, in contrast to the atomic case. It is such that we

cannot take a once-and-for-all fixed single-particle potential and hope to find the corresponding single-particle states realized very accurately, be it even over a very limited range of neighboring nuclei.

We should keep these precautions in mind when talking about the shell model. As we said, the filling of the shells is without ambiguity, if we have closed shells. When we start filling neutrons and protons in *unfilled shells* these states will be degenerate, because the j-shells have a $(2j+1)$-fold degeneracy. The configuration of the nucleus can then be characterized by two numbers, κ and λ, which stand for the proton and neutron numbers, respectively, in the partially filled j-shell. Let the partially filled neutron shell be characterized by the quantum numbers $(n\ l\ j)$, and the partially filled proton shell by $(n'\ l'\ j')$. One then denotes the configuration by

$$(\nu nlj)^{\kappa}(\pi n' l' j')^{\lambda}.$$

Because of the $2j+1$-fold degeneracy of each j-shell, all possible shell model states corresponding to this configuration are also degenerate. The number of antisymmetric, linearly independent products is given by the product of the binomial coefficients

$$\binom{2j+1}{\kappa}\binom{2j'+1}{\lambda}. \tag{2.37}$$

The degeneracy of all these states will, of course, be removed in reality due to the action of the residual interaction V_R (2.35), which is neglected in the shell model. Taking one of the phenomenological nucleon–nucleon forces, as discussed in Chapter 4, one can diagonalize V_R in the subspace (2.37). Usually one takes not only this subspace into account, but also the one which corresponds to the nearly degenerate levels of a whole major shell. The $s_{1/2}$, $d_{5/2}$, $d_{3/2}$ levels of the $s-d$ shell is such a case, covering nuclei from ^{16}O up to ^{40}Ca. One can easily be convinced that the dimension of the matrices to be diagonalized becomes exceedingly large for more than two particles in open shells. Special procedures have been developed to diagonalize such huge matrices [Wh 72, SZ 72, WWC 77].

To reduce the size of these matrices, symmetries such as isospin or angular momentum (see Sec. 2.6) can be of great help (see, for instance, [FHM 69, WMH 71, HMW 71, GED 71, VGB 72, Wi 76]).

2.6 Symmetry Properties

2.6.1 Translational Symmetry

For any solution of the eigenvalue problem (2.19) we must require that a series of symmetry or invariance properties are fulfilled. Among these are, for example, translational and rotational invariance.* In the shell model

* Besides these exact symmetries, in some regions of the periodic table one often also has approximate symmetries, as, for instance, the isospin (see Sec. 2.6.3), which can be used for a classification of spectra (see [He 73a]).

one of these invariances is always violated: the *translational invariance*. This comes from the fact that we have to fix the potential in space in contradiction to the homogeneity of space. The most serious consequence of this violation is the appearance of spurious states in the excitation spectrum of the system. This occurs because we have introduced redundant degrees of freedom. If we fix the nucleus in space, it has only $3A-3$ spatial degrees of freedom left. The shell model, however, contains $3A$ variables. These spurious states are therefore not true excitations of the system, but correspond to motions of the nucleus as a whole. Almost all approximation schemes in nuclear physics have inherent symmetry violations. In Chapter 11 we will, therefore, show in fair detail how such violations can be removed.

2.6.2 Rotational Symmetry

The spherical shell model Hamiltonian H_0 (**ls** term included) conserves rotational symmetry. Therefore, it is possible to construct eigenstates of the total angular momentum

$$\mathbf{J}=\sum_{i=1}^{A}\mathbf{j}^{(i)} \quad \text{and} \quad J_z=\sum_{i=1}^{A} j_z^{(i)} \tag{2.38}$$

by a linear combination of the Slater determinants (2.23). The closed shell ground state is not degenerate; the only nondegenerate angular momentum eigenstate has $I=0$, which is therefore identified with the ground state. This consequence is experimentally confirmed with no exception. It is then clear that, having only one nucleon outside closed shells, the ground state angular momentum of such even–odd nuclei will correspond to the j-value of the odd nucleon. The same, of course, is true if there is one nucleon missing (a hole) in a filled shell. This rule is also experimentally confirmed with only very few exceptions.

If we fill (remove) more than one particle into (from) an unfilled (filled) j-level, the situation gets more complicated, because different I-values will be degenerate. Again, we can remove this degeneracy by diagonalizing the residual interaction. The matrices are now much smaller as we get one for each I value.

The construction of eigenfunctions of $\mathbf{J}^2$ will be shown explicitly for a very simple example (for more complicated situations, see [ST 63]). Suppose that in a j-shell there are only two protons, the configuration of which is then $(\pi j)^2$. Out of the degenerate two-particle states (which we want to denote by $|m_1 m_2\rangle$, m being the magnetic quantum number), we construct, according to the rules of angular momentum coupling, an eigenstate $|IM\rangle$ of $\mathbf{J}^2$ and J_z with

$$\mathbf{J}^2=(\mathbf{j}_1+\mathbf{j}_2)^2. \tag{2.39}$$

We obtain

$$|IM\rangle=\frac{1}{\sqrt{2}}\sum_{m_1 m_2} C^{j\ j\ I}_{m_1\ m_2\ M}|m_1 m_2\rangle. \tag{2.40}$$

Using the symmetry properties of the Clebsch–Gordan coefficients, (see [Ed 57, Eq. (3.5.14)]), we have with proper normalization

$$|IM\rangle = \frac{1}{\sqrt{8}} \sum_{m_1 m_2} \left(C^{j}_{m_1\, j\, m_2\, I\, M} + (-)^{2j-I} C^{j}_{m_2\, j\, m_1\, I\, M} \right) |m_1 m_2\rangle \tag{2.41}$$

which, upon noting the antisymmetry $|m_2 m_1\rangle = -|m_1 m_2\rangle$ yields

$$|IM\rangle = \frac{1}{\sqrt{8}} \sum_{m_1 m_2} C^{j}_{m_1\, j\, m_2\, I\, M} \left(1 + (-)^{2j-I+1}\right) |m_1 m_2\rangle. \tag{2.42}$$

We see from Eq. (2.42) that $|IM\rangle$ is only different from zero for

$$2j - I + 1 = 2n \quad \text{or} \quad I = 2n; \qquad n = 0, 1, 2\ldots, \tag{2.43}$$

that is , for even angular momenta. Taking as a definite example $j = 3/2$, we see that the six independent components

$$|m_1 m_2\rangle : |\tfrac{3}{2}\tfrac{1}{2}\rangle, |\tfrac{3}{2} -\tfrac{1}{2}\rangle, |\tfrac{3}{2} -\tfrac{3}{2}\rangle, |\tfrac{1}{2} -\tfrac{1}{2}\rangle, |\tfrac{1}{2} -\tfrac{3}{2}\rangle, |-\tfrac{1}{2} -\tfrac{3}{2}\rangle$$

have been transformed by a unitary transformation to the six angular momentum coupled components, which are degenerate among themselves:

$$|IM\rangle : |00\rangle, |22\rangle, |21\rangle, |20\rangle, |2-1\rangle, |2-2\rangle.$$

In the general case these considerations are a little tedious. In Table 2.1 the possible total angular momenta for a pure proton configuration are presented.

The factor $1/\sqrt{2}$ in Eq. (2.40) is a normalization in the case in which both particles are in the same shell. In general, we have for the coupling of two particles:

$$(a^{+}_{nlj} a^{+}_{n'l'j'})_{JM} = \frac{1}{\sqrt{1 + \delta_{jj'}\delta_{ll'}\delta_{nn'}}} \sum_{mm'} C^{j}_{m\, j'\, m'\, J\, M} a^{+}_{nljm} a^{+}_{n'l'j'm'}. \tag{2.44}$$

Special care has to be applied in coupling hole states. The operator a^{+}_{nljm} transforms like the eigenfunction ϕ_{nljm}, that is cogredient, under a rotation of the coordinate system and therefore like a spherical tensor of rank j (i.e., with $D^{j}_{mm'}$; see Appendix A). On the other hand, the annihilation operator a_{nljm} transforms with $D^{j*}_{mm'}$, that is, contragredient. The normal coupling rules (2.44) apply only for tensors, which are both cogredient or both contragredient. We can therefore only couple a^{+}_{nljm} with the time reversed operator (see [Me 61]),

$$a_{\overline{nljm}} = T a_{nljm} T^{+} = (-)^{l+j-m} a_{nlj-m}, \tag{2.45}$$

which is, of course, cogredient. The *ph* coupling rule is therefore

$$(a^{+}_{nlj} a_{n'l'j'})_{JM} = \sum_{mm'} (-)^{j'-m'} C^{j}_{m\ -m'\ J\, M} a^{+}_{nljm} a_{n'l'j'm'}, \tag{2.46}$$

where we have left out the unimportant phase $(-)^{l'+1}$.

From pure angular momentum coupling one cannot as yet decide which of the degenerate states corresponds to the ground state. For that we have, as we have said, to diagonalize V_R in a certain subspace. This confirms, in general, the experimentally observed rule that even–even groundstates have $I = 0$.

Another experimentally found coupling rule which the pure shell model cannot explain without configuration mixing is the fact that even odd nuclei far from closed shells have ground state spins equal to the j-value of the odd particle. We will see in Chapter 6 how this finds a natural explanation by taking correlations of the nucleons into account.

Table 2.1 List of possible total angular momenta I for the configuration $(j)^\kappa$ ([MJ 55, p. 64])

	κ	
$j=\frac{1}{2}$		
	1	$\frac{1}{2}$
$j=\frac{3}{2}$		
	1	$\frac{3}{2}$
	2	0, 2
$j=\frac{5}{2}$		
	1	$\frac{5}{2}$
	2	0, 2, 4
	3	$\frac{3}{2}, \frac{5}{2}, \frac{9}{2}$
$j=\frac{7}{2}$		
	1	$\frac{7}{2}$
	2	0, 2, 4, 6
	3	$\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}, \frac{15}{2}$
	4	0, 2 (twice), 4 (twice), 5, 6, 8
$j=\frac{9}{2}$		
	1	$\frac{9}{2}$
	2	0, 2, 4, 6, 8
	3	$\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}$, (twice), $\frac{11}{2}, \frac{15}{2}, \frac{17}{2}, \frac{21}{2}$
	4	0 (twice), 2 (twice), 3, 4 (3 times), 5, 6 (3 times), 7, 8 (twice), 9, 10, 12
	5	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ (twice), $\frac{7}{2}$ (twice), $\frac{9}{2}$ (3 times), $\frac{11}{2}$ (twice), $\frac{13}{2}$ (twice), $\frac{15}{2}$ (twice), $\frac{17}{2}$ (twice), $\frac{19}{2}, \frac{21}{2}, \frac{25}{2}$
$j=\frac{11}{2}$		
	1	$\frac{11}{2}$
	2	0, 2, 4, 6, 8, 10
	3	$\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}$ (twice), $\frac{11}{2}$ (twice), $\frac{13}{2}, \frac{15}{2}$ (twice), $\frac{17}{2}, \frac{19}{2}, \frac{21}{2}, \frac{23}{2}, \frac{27}{2}$
	4	0 (twice), 2 (3 times), 3, 4 (4 times), 5 (twice), 6 (4 times), 8 (4 times), 9 (twice), 10 (3 times), 11, 12 (twice), 13, 14, 16
	5	$\frac{1}{2}, \frac{3}{2}$ (twice), $\frac{5}{2}$ (3 times), $\frac{7}{2}$ (4 times), $\frac{9}{7}$ (4 times), $\frac{11}{2}$ (5 times), $\frac{13}{2}$ (4 times), $\frac{15}{2}$ (5 times), $\frac{17}{2}$ (4 times), $\frac{19}{2}$ (4 times), $\frac{21}{2}$ (3 times), $\frac{23}{2}$ (3 times), $\frac{25}{2}$ (twice), $\frac{27}{2}$ (twice), $\frac{29}{2}, \frac{31}{2}, \frac{35}{2}$
	6	0 (3 times), 2 (4 times), 3 (3 times), 4 (6 times), 5 (3 times), 6 (7 times), 7 (4 times), 8 (6 times), 9 (4 times), 10 (5 times), 11 (twice), 12 (4 times), 13 (twice), 14 (twice), 15, 16, 18

2.6.3 The Isotopic Spin

Up to now we have always considered the neutrons and protons separately. As a consequence we have, for example, in Eq. (2.23), a product of two Slater determinants—one for protons and one for neutrons. Apart from their electromagnetic interactions, protons and neutrons have practically the same physical properties. We will see, for instance, in Chapter 4, that nuclear forces are to a large extent independent of whether we consider protons or neutrons—that is, they are charge independent. As long as the influence of the Coulomb force on the nuclear properties can be neglected, we can consider the proton and the neutron as just two

different manifestations of the same particle: the nucleon. Mathematically we say that the nucleon can be in two different states, the basis vectors of which may be written as $\pi = \binom{0}{1}$ and $\nu = \binom{1}{0}$, thus forming a two-dimensional space. This then is very similar to what we have in the spin case. Instead of a quantum number indicating whether the nucleon spin is up or down, we now have an additional quantum number, indicating whether the nucleon is a proton or a neutron, called the isotopic spin because of its formal analogy to ordinary spin.* With this additional quantum number we can write the nuclear shell model wave function as a single determinant. However one should realize that, contrary to the ordinary spin, the isospin has nothing to do with rotations in the coordinate space. Therefore, the treatment of isopsin is much simpler than that of ordinary spin which, unlike the isospin, has to be coupled to the orbital angular momentum.

As in the case of ordinary spin, one can set up the usual spin algebra:

$$t_3 \nu = \tfrac{1}{2}\nu, \qquad t_3 \pi = -\tfrac{1}{2}\pi, \tag{2.47}$$

with

$$t_3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.48}$$

The lowering and raising operators t_- and t_+ change the neutron into a proton and vice versa, respectively,

$$t_- \nu = \pi, \quad t_+ \nu = 0, \qquad t_- \pi = 0, \quad t_+ \pi = \nu, \tag{2.49}$$

with

$$t_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \qquad t_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{2.50}$$

Like ordinary spin, the isospin vector operator is formed out of its three cartesian components

$$\mathbf{t} = \{t_1, t_2, t_3\}. \tag{2.51}$$

We can therefore define the total isospin of a system of A nucleons

$$\mathbf{T} = \sum_{i=1}^{A} \mathbf{t}^{(i)} \tag{2.52}$$

and its 3-component

$$T_3 = \sum_{i=1}^{A} t_3^{(i)}. \tag{2.53}$$

Since we have the same algebra for isotopic spin as for ordinary spin, the same coupling rules can be applied.

The 3-component of the total isospin is a measure of the total neutron

*The concept of isospin was introduced originally by Heisenberg [He 32]. Later on it was much developed by Wigner [Wi 37].

excess of the nucleus:

$$T_3\Psi(1\ldots A)=\{-\tfrac{1}{2}-\tfrac{1}{2}\cdots-\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2}\cdots+\tfrac{1}{2}\}\Psi(1\ldots A)$$
$$=\tfrac{1}{2}(N-Z)\Psi(1\ldots A). \tag{2.54}$$

The eigenvalue is therefore equal to half the neutron excess. For systems where $\mathbf{T}^2$ commutes with the nuclear Hamiltonian, the eigenvalue T corresponding to $\mathbf{T}^2$ can take on the values

$$|T_3| \leqslant T \leqslant \frac{A}{2}\,. \tag{2.55}$$

Each nuclear state then has a good total isospin quantum number T as long as Coulomb forces, which do not commute with $\mathbf{T}^2$, can be neglected.

As a simple example, we will show how to attribute total isotopic spin to certain nuclear states. For the three isobar nuclei

$$^{10}_{4}\mathrm{Be}_6, \qquad ^{10}_{5}\mathrm{B}_5, \qquad ^{10}_{6}\mathrm{C}_4,$$

the 3-component of the total isospin is, according to Eq. (2.54),

$$T_3^{(\mathrm{Be})}=-1, \qquad T_3^{(\mathrm{B})}=0, \qquad T_3^{(\mathrm{C})}=1. \tag{2.56}$$

The total isospin can therefore be either $T=1$ or $T=0$ [Eq. (2.55)].

In the first case we speak of an isospin triplet, states of which should be found in all three nuclei at the same energy (slight differences will nevertheless occur because of the Coulomb energy); the second case is an isospin singlet, which can only be attributed to boron. In Fig. 2.9 we show the low-lying states of the three isobars and their identification with isospin quantum numbers.

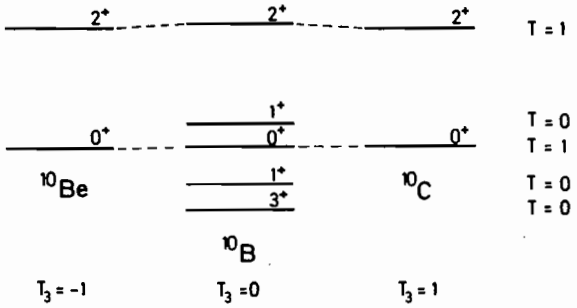


Figure 2.9. Isospin triplets and singlets in the excitation spectrum of isobars.

From these considerations it may seem that the concept of isospin can only be applied to light nuclei, where the Coulomb energy is negligible. However it turns out in *heavy nuclei* also, that the low-lying states have almost good isospin. This can be understood qualitatively using the following arguments.

We first consider a pure shell model and assume that the wave functions for protons and neutrons with the same quantum numbers are identical, that is that the proton and the neutron wells are identical to within a constant shift Δ_c (2.27). This is a rather good approximation for the bound states.

In Fig. 2.10 we have side by side the levels with the same quantum numbers. This is a bit misleading for energy arguments, but useful in the following discussion.

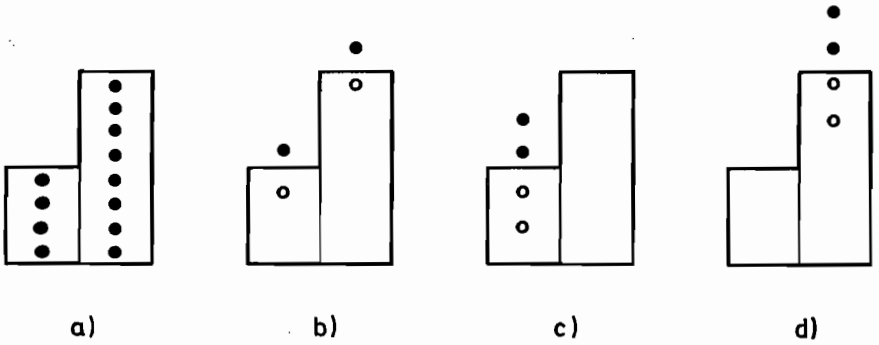


Figure 2.10. Schematic representation of shell model configurations.

From Fig. 2.10a we see that a shell model state where all occupied proton levels are also occupied by neutrons has $T = T_3$, because operating with $T_+ = \sum_i t_+^{(i)}$ onto this state gives 0, that is, this state has the maximal value of T_3 for a fixed value of T. This shows in particular that the shell model ground state for $N = Z$ nuclei has $T = 0$.

For non-magic heavy nuclei the shell model ground state is certainly not a good approximation to the exact ground state. There are admixtures of $1p$-$1h$, $2p$-$2h$ states. As we will show in Chapter 5, the construction of a self-consistent, one-particle potential assures that there are no $1p$-$1h$ admixtures. Since the shell model potential is a good approximation to such a potential, we neglect these contributions here and argue about $2p$-$2h$ admixtures. Furthermore, we assume that $\Delta_\epsilon \simeq \hbar\omega_0$, which is the case for heavy nuclei like ^{208}Pb, and neglect $2p$-$2h$ configurations with an unperturbed energy larger than $2\hbar\omega_0$. As we see from Fig. 2.10b, c, d there are three types of configurations. Only type b contains configurations which violate the isospin, namely, those where the proton particle sits in the same level as the neutron hole. The ratio of the number of these configurations to the number of all configurations of type b is $1/(N-Z)$, where $(N-Z)$ is the neutron excess. If we make the statistical assumption that all these configurations are admixed with the same weight, we find that the isospin impurity in the ground state of heavy nuclei is roughly $1/(N-Z)$. Similar considerations apply for excited states, and we find that heavy nuclei with large neutron excesses have rather pure isospin. This result has also been found in more detailed investigations [LS 62, SK 65, KW 69, So 69, LM 74].

2.7 Comparison with Experiment

2.7.1 Experimental Evidence for Single-Particle (Hole) States

Besides the success of the shell model as an explanation of the magic numbers and angular momenta of the ground state, one would like to have direct experimental evidence for the nuclear shells. It turns out that direct stripping [e.g., (d, p)] and pickup (p, d) reactions, as well as direct $(p, 2p)$ and $(e, e'p)$ reactions, are well suited for this purpose. For example, a direct $(p, 2p)$ process is ideally one where the energy of the incoming proton is so high that it interacts only once with another proton in the

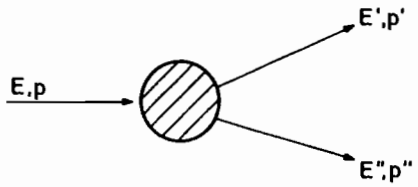


Figure 2.11. Schematic representation of the $(p, 2p)$ reaction.

nucleus, both protons leaving the nucleus without further interactions (Fig. 2.11).

Suppose that the knocked-out nucleon has been in a shell model state (Fig. 2.12). For its binding energy we have

$$E_B = E - E' - E''.$$

The cross section according to the shell model should have, as a function of the energy, resonances only at discrete values of E_B.

In reality, these resonances are broadened because of the influence of the residual interaction, as shown in Fig. 2.13 for the experimental cross section of ^{16}O $(p, 2p)$ ^{15}N. Nevertheless, one can clearly identify the

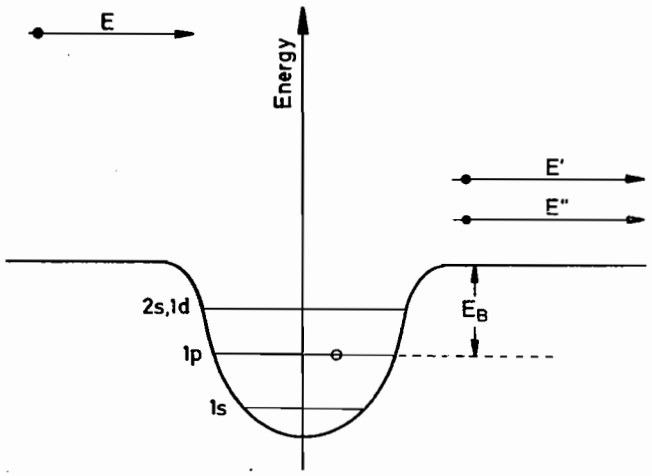


Figure 2.12. $(p, 2p)$ reaction in the shell model.

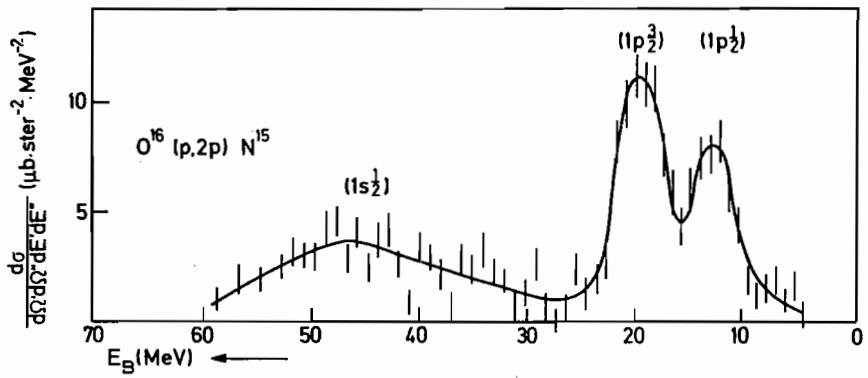


Figure 2.13. Experimental ^{16}O $(p, 2p)$ ^{15}N cross section [MHT 58].

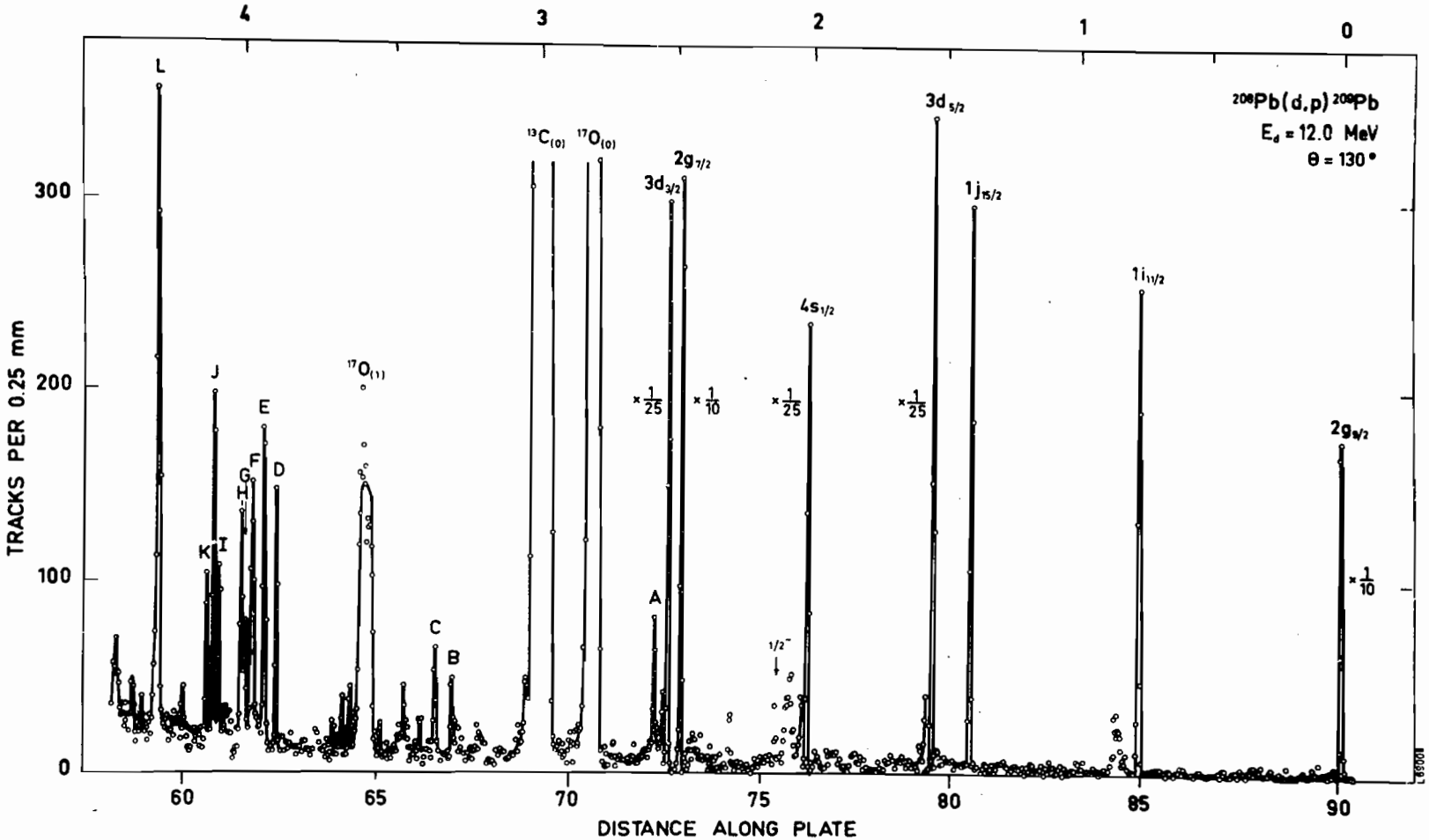


Figure 2.14. Neutron spectrum from the ^{208}Pb (d, p) ^{209}Pb reaction. The intensity peaks are labelled by their shell model assignment. (From[EKV 69]

($\pi 1s_{1/2}$) level and the spin orbit split levels ($\pi 1p_{3/2}$) and ($\pi 1p_{1/2}$). Theoretical calculations for these reactions have been done by R. Lipperheide et al. [FLW 75].

In a similar manner, one can use stripping or pickup reactions like (d, p) and (p, d) to create neutron particle or neutron hole states, but the kinematical analysis has to be corrected for the binding of the deuteron. A nice example for neutron particle states in ^{209}Pb obtained with a (d, p) stripping reaction is presented in Fig. 2.14. It should be compared with Fig. 2.15, where experimentally determined proton and neutron single-particle energies for ^{208}Pb are shown in comparison with calculated levels, using a Woods–Saxon potential [BW 60].

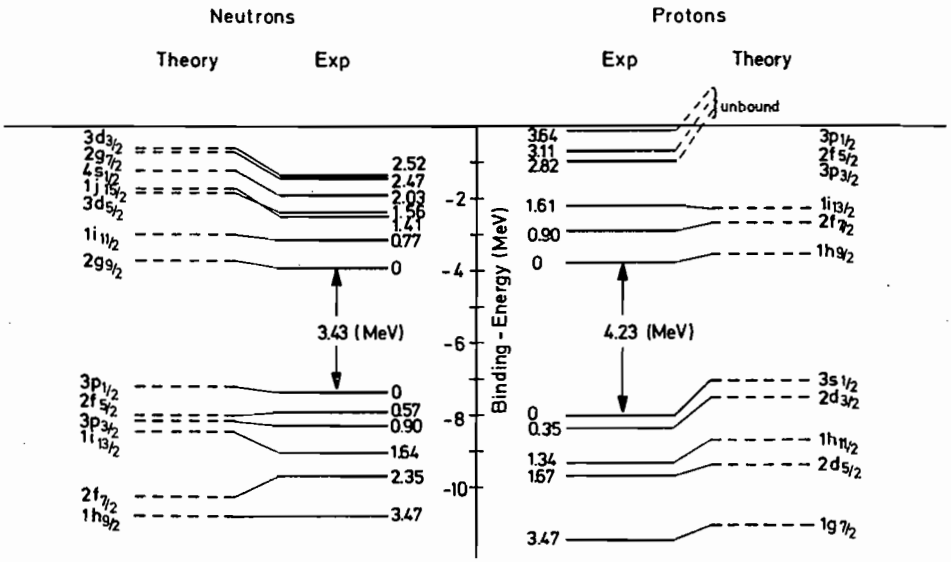


Figure 2.15. Comparison of experimental and calculated single-particle levels in ^{208}Pb.

One can show (see, for instance, [GL 70, Au 70, Ho 71b, EW 72]) that the absolute values of the cross sections of these reactions are proportional to the so-called spectroscopic factor

$$S_k = |\langle \Psi_{A+1} | a_k^+ | \Psi_A \rangle|^2, \tag{2.57}$$

where $|\Psi_A\rangle$, $|\Psi_{A+1}\rangle$ are the exact wave functions of the initial and final nucleus, and k stands for the quantum numbers of the observed single-particle state. If for the wave functions we use the pure shell model approximation, the spectroscopic factor is clearly equal to one.* However, if the wave functions contain admixtures of more complicated particle and hole configurations, which are introduced by the diagonalization of the residual interaction (2.35), the spectroscopic factor differs from one. Therefore, it is a measure of the purity of the single-particle states. The spectroscopic factors of the levels shown in Fig. 2.14 are in the range 0.8–0.9 (see, for instance, [RW 73]).

*This statement applies only for pure product states without angular momentum coupling. With angular momentum coupling, the different levels with different m values are partially filled and one has to take into account the Pauli principle.

2.7.2 Electromagnetic Moments and Transitions

Other important quantities ready for comparison with experiments are the electromagnetic moments and transition probabilities in odd mass nuclei. In a first approximation, one assumes that they are described by one external particle sitting on an inert core with even mass:

$$|\Phi_m\rangle = a_m^+ |\Phi_0\rangle, \tag{2.58}$$

where a_m^+ creates a particle in the level m and $|\Phi_0\rangle$ represents the wave function of the core (2.25).

The electromagnetic multipole operators $\hat{Q}_{\lambda\mu}$ and $\hat{M}_{\lambda\mu}$ (as defined in Appendix B), are one-particle operators. As shown in Appendix C, they can be represented as

$$\hat{Q} = \sum_{k_1 k_2} Q_{k_1 k_2} a_{k_1}^+ a_{k_2} \tag{2.59}$$

in the shell model basis. Using the Fermi commutation relations (C.23) for the operators a_k, a_k^+ and the fact that the levels m, m' are empty in the core ($a_m|\Phi_0\rangle = a_{m'}|\Phi_0\rangle = 0$), we get

$$\langle\Phi_m|\hat{Q}|\Phi_{m'}\rangle = \langle\Phi_0|\hat{Q}|\Phi_0\rangle\delta_{mm'} + Q_{mm'}. \tag{2.60}$$

The picture becomes especially simple if we assume that the core is a closed shell nucleus. In this case, all the electromagnetic matrix elements of the core ($I=0$) vanish. The single-particle states are characterized by the quantum numbers $|k> = |nljm>$ and the electromagentic properties can be calculated from the matrix elements

$$Q_{kk'} = \langle nljm|\hat{Q}|n'l'j'm'\rangle, \tag{2.61}$$

which are given in Eqs. (B.81) and (B.82) of Appendix B.

For instance, we get the electromagnetic moments [(B.31) and (B.32)] as expectation values in the states $|nljm=j\rangle$ and the electromagnetic transition probabilities (B.72) from the reduced matrix elements of these operators.

Special care has to be taken in the case of single hole states (one particle less than a closed shell). As we have seen, they can be described in the same simple way as the single-particle states. Their wave function can be expressed by analogy with Eq. (2.58):

$$|\Phi_i\rangle = a_i|\Phi_0\rangle, \tag{2.62}$$

where $|\Phi_0\rangle$ is again the wave function of the core and a_i annihilates a particle in a level i occupied in $|\Phi_0\rangle$. By analogy with (2.60), we get

$$\langle\Phi_i|\hat{Q}|\Phi_{i'}\rangle = \langle\Phi_0|\hat{Q}|\Phi_0\rangle\delta_{ii'} - Q_{i'i}. \tag{2.63}$$

Since the electromagnetic multipole operators are self-adjoint in the sense of [Ed 57, Eq. (5.5.2)], the difference between (2.63) and (2.60) does not enter into the electromagnetic transition probabilities. However, in the case of electromagnetic moments, one has to take care of the sign in (2.63) and an additional phase. Since one is interested in the expectation value of a

state with angular momentum quantum numbers $j, m=j$, one has to create a hole in the level $|jm=-j\rangle$, which is proportional to the time-reversed level $T|jm=j\rangle$ [Eq. (2.45)]. Together with the minus sign in Eq. (2.63), one finds that hole states have the same magnetic moments as the corresponding particle states (the magnetic multipole operators are time odd), whereas the electric multipole moments of particle and hole states have different signs (the electric multipole operators are time even).

One example is the magnetic dipole moment (B.31)

$$\mu=\langle nljj|\,\mu_z|nljj\rangle=\sqrt{\tfrac{4\pi}{3}}\,\langle nljj|\hat{M}_{10}|nljj\rangle, \tag{2.64}$$

where the magnetic vector $\boldsymbol{\mu}$ is given by

$$\boldsymbol{\mu}=\mu_N\{g^l\mathbf{l}+g^s\mathbf{s}\}, \tag{2.65}$$

where $\mu_N=e\hbar/2mc$ is the nuclear magneton and g^l and g^s are the gyromagnetic rations for orbital angular momentum and spin ($g^l=1$, $g^s=5.586$ for protons and $g^l=0$, $g^s=-3.826$ for neutrons).

The magnetic moment μ can easily be calculated with the projection theorem for vector operators $\mathbf{A}$,

$$\langle jm|\mathbf{A}|jm'\rangle=\langle jm|\mathbf{j}|jm'\rangle\frac{\langle jj|\mathbf{Aj}|jj\rangle}{j(j+1)}, \tag{2.66}$$

which can be derived from the Wigner–Eckart theorem [Ed 57, Eq. (5.4.1)]. Applying it to the vector $\boldsymbol{\mu}$, we get

$$\begin{aligned}\mu&=\mu_N\frac{1}{j+1}\langle jj|\,g^l\mathbf{lj}+g^s\mathbf{sj}|jj\rangle\\&=\mu_N\frac{1}{2(j+1)}\left[g^l(j(j+1)+l(l+1)-\tfrac{3}{4})+g^s(j(j+1)+\tfrac{3}{4}-l(l+1))\right].\end{aligned} \tag{2.67}$$

For $j=l\pm\frac{1}{2}$ we get

$$\mu=\mu_N\begin{Bmatrix}g^l(j-\frac{1}{2})+\frac{1}{2}g^s\\\left[g^l(j+\frac{3}{2})-\frac{1}{2}g^s\right]\frac{j}{j+1}\end{Bmatrix}\quad\text{for}\quad\begin{cases}j=l+\frac{1}{2}\\j=l-\frac{1}{2}\end{cases}. \tag{2.68}$$

The functions $\mu(j)/\mu_N$ for $j=l\pm 1/2$ are called Schmidt lines (Figs. 2.16 and 2.17) (j can only take on discrete values $\mu(j)/\mu_N$, though it is shown here as a continuous function of j). The experimental values are given for comparison.

If the theory were exact, all μ-values would lie on the Schmidt lines and all orbital angular momenta $l=j\pm\frac{1}{2}$ would be determined. The experimental values, almost without exception, lie in between the two lines, most of them being grouped closer to one of the two lines. The l-values determined in this way almost always agree with those predicted by the shell model. The shell model should work best near magic numbers. Indeed, one finds close agreement, for example, for ^{15}N, ^{17}O, ^{39}K, ^{41}K, and ^{207}Pb. The

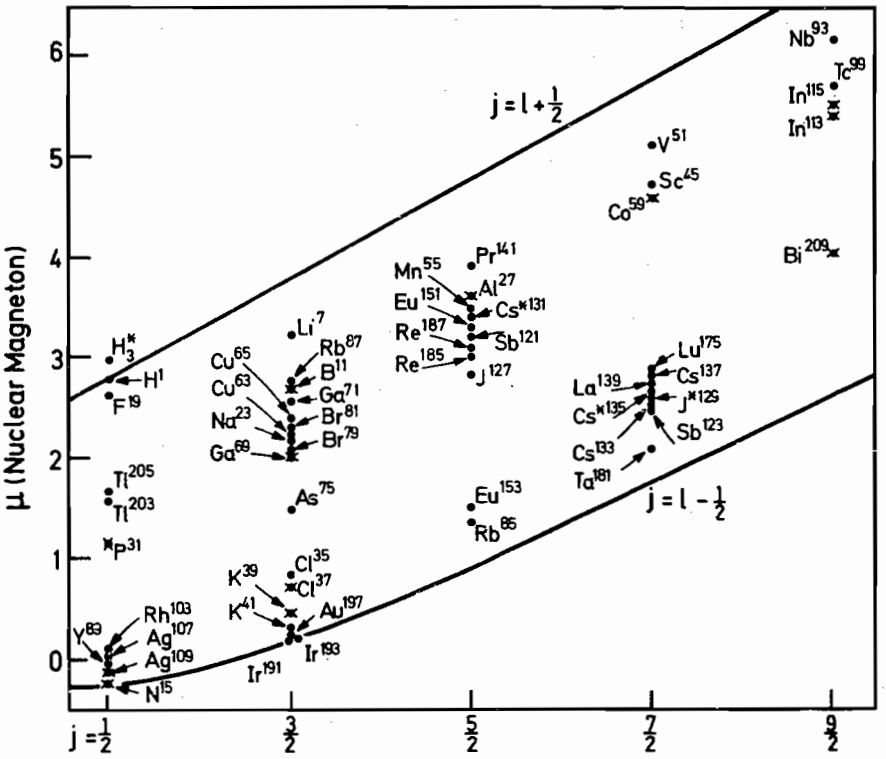


Figure 2.16. Magnetic moments for Z-odd nuclei as a function of angular momentum. (From [MJ 55].)

agreement is also not too bad for many nuclei far from closed shells. One can qualitatively understand this, because the magnetic moment in time-reversed levels is always of opposite sign. Since, as we shall see in Chapter 6, time-reversed levels are always occupied pair-wise, the most important contribution comes from the last odd particle. This is, however, only a

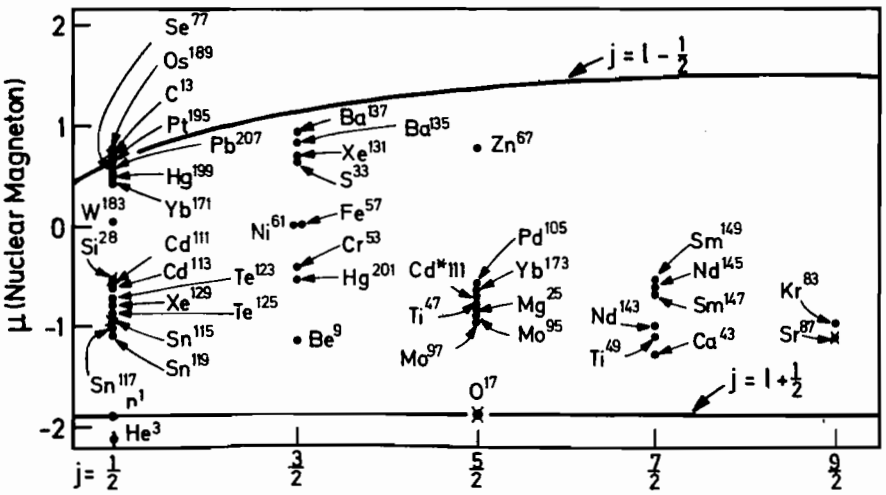


Figure 2.17. Magnetic moments of N-odd nuclei as a function of angular momentum. (From [MJ 55].)

qualitative argument and one also finds many deviations, especially for heavier nuclei, even in cases of rather pure one-particle or one-hole configurations.

There are two main reasons for these deviations:

(i) As we see in Appendix B, we have used the g-factors of bare nucleons in the calculation of the magnetic multipole operators. Within the nucleus however, there exist pionic currents between the nucleons that change these values, and effective g-factors, $\tilde{g}$, should be used.

(ii) The core usually does not stay inert in an external electromagnetic field. This can excite virtual vibrations which interact with the external particle. This effect is called a polarization of the core. To a large extent, it can be taken into account again by effective g-values. We will see in Section 9.3 how this polarization effect can be calculated microscopically.

The most important magnetic transitions are M1-transitions. In the pure single-particle model, the M1-operator (2.65) commutes with $\mathbf{l}^2$. This means that M1-transitions between levels with different l-quantum numbers are forbidden (l-forbidden transitions). In fact, such transitions are observed experimentally, but with a very small BM1-value. They cannot be understood by using only effective g-values, because one does not change the selection rules in this case. This clearly shows that mesonic effects (i) and polarizations (ii) produce not only vector components $\sim\mathbf{l}$ and $\sim\mathbf{s}$, but also more complicated effects. The simplest one is a "tensor component"

$$\delta\boldsymbol{\mu} = \kappa r^2 [\, Y_2 \mathbf{s}\,]_1 t_3 . \tag{2.69}$$

By adjusting the constant κ in a reasonable way, one is able to describe these l-forbidden transitions quantitatively (see [BM 69, WB 69, and BSK 73]).

Let us come now to the electric properties. There are two important differences to the magnetic case:

(i) The electric operators are time-even. There are no cancellations of the contributions of time-reversed levels, but on the contrary they add up. This means that one cannot expect that the spherical single-particle model works outside the region of closed shells.

(ii) Since one can, in the long wavelength limit, (see Appendix B) neglect the contributions of currents to the electric multipole operators, we can use the conservation law for the electric charge and show that exchange effects do not play any role in the electric multipole operators. This means that apart from polarization effects, we can use the bare electric charge (*Siegert theorem* [Si 37]).

By analogy with the magnetic moment, one obtains for the electric quadrupole moment (B.81) of a single-particle state [EL 57, p. 255]

$$Q = e\sqrt{\frac{16\pi}{5}}\ \langle nljj|r^2 Y_{20}|nljj\rangle = -e\langle r^2\rangle \frac{2j-1}{2j+2}, \tag{2.70}$$

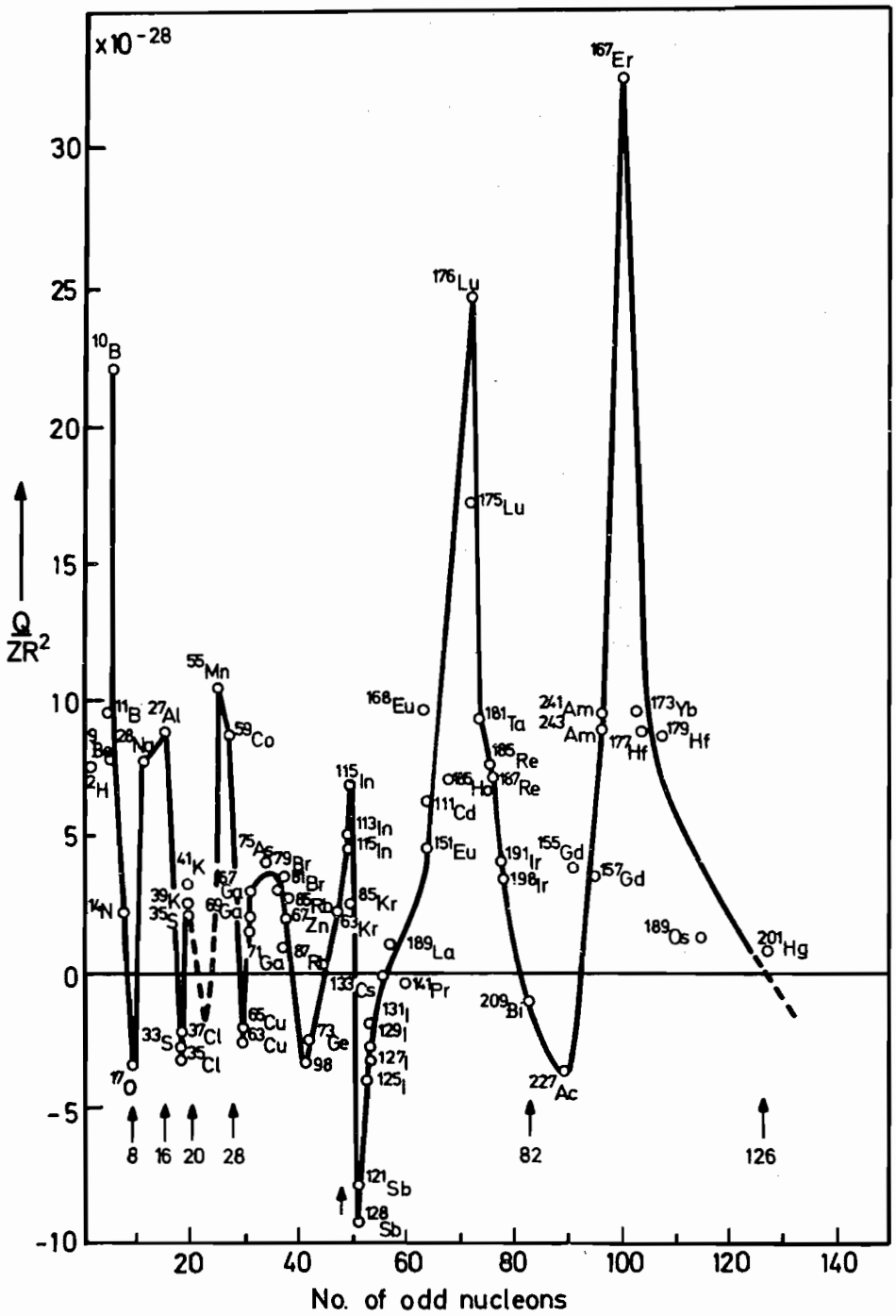


Figure 2.18. Quadrupole moments for odd-A nuclei against the number of odd nucleons. Arrows indicate closed shells. (From [Se 64].)

where $\langle r^2 \rangle$ is the average square radius of a particle in the state $|nljm\rangle$. For $j > 1/2$, the quadrupole moments turn out to be negative, that is, the probability distribution looks like a pancake in the plane perpendicular to the z-axis. For holes we have the opposite (i.e., prolate probability distributions). In Fig. 2.18, we see that this picture is qualitatively right in the neighborhood of magic numbers. However, as one fills the next shell with more and more nucleons, we soon find experimentally a transition to positive quadrupole moments with very large values. We shall see in the next chapter that in this case the average field of the nucleons is no longer spherical and we obtain a deformed density distribution. Only at the end of the shell, when nearly all levels are occupied, does one again get the picture of one or a few holes in a spherical core.

Even for one-particle or one-hole states with a magic core, this picture only gives qualitative agreement. Quantitatively, the measured quadrupole moments for proton states are roughly a factor of two larger than one would expect from (2.70) and values for neutron states do not vanish as they should according to Eq. (2.70), but behave as if the neutron had a charge. One usually expresses this fact by an effective charge e^{eff}. These effective charges can be explained by the polarization effect (see Sec. 9.3). Experimentally, it is observed that $e^{\text{eff}} \simeq 1$ for neutrons and $e^{\text{eff}} \simeq 2$ for protons.

The electric transitions behave very much like the corresponding moments. Only for the single-particle and single-hole nuclei near closed shells do they have values predicted by the single-particle shell model, with roughly the same effective charge as determined from the quadrupole moments.

Levels with *high angular momenta near closed shell* nuclei are often a mixture of only very few configurations (often there is only one configuration possible in a wide energy range). Sometimes—because of the selection rules of spin and parity—they can decay only by radiation with a high multipolarity. From Appendix B we learn that such transitions are highly suppressed because of kinematical factors. We therefore expect a very long lifetime for such nuclei. In fact, quite a few such "isomeric" states have been found in spherical nuclei (islands of isomers; see also Sec. 3.4.7).

2.8 Deformed Shell Model

2.8.1 Experimental Evidence

The assumption of approximately independent motion of nucleons in an average field is the basis of the shell model and of all microscopic theories of finite nuclei. This container potential is produced by the nucleons themselves and their mutual interaction. In Chapter 5 we shall see how to calculate this average potential. In its most simple form the potential well is spherical. This is true for nuclei with closed or nearly closed shells and, as we have seen in the last section, for such nuclei the spherical shell model

is very successful. Far from closed shells—that is, for mass numbers $A \simeq 25$ (Al, Mg), $150 < A < 190$ (nuclei of the rare earths) and for $A > 220$ (the actinides)—the independent particle picture again works quite well. However, in these regions one has to assume a deformed single particle potential. [Ra 50, 76, Bo 51].

The assumption of a deformation is able to explain many experimental facts, of which the most important are:

(i) **The existence of rotational bands.** In the mass regions mentioned above, the nuclear excitation spectra show pronounced rotational bands with an $I(I+1)$ spectrum, as in Eq. (1.64) (see, for instance, [BM 75]). As discussed in Chapter 1.5, such collective rotational bands are closely related to stable nuclear deformations.

(ii) **Very large quadrupole moments.** We have already seen in Section 2.7 that the spherical single-particle model with an inert core is by no means able to explain the large quadrupole moments in the regions far from closed shells (Fig. 2.18). This experimental fact is a hint about stable nuclear deformations, where the core also contributes to the quadrupole moment. Using the Bohr model [Eq. (1.75)] we can determine from the experimental (spectroscopic) quadrupole moment Q an intrinsic value Q_0, and derive a deformation parameter β [Eqs. (1.13) and (1.72)]. The "experimental" values of β, determined in this way, are for axial symmetric nuclei ($\gamma = 0$) in the rare earth region around

$$\beta \simeq 0.2\text{–}0.3. \tag{2.71}$$

The sign is positive. This means that we have cigar-shaped or prolate deformations. In other regions (for instance, $A \sim 25$, $A \sim 150$ or $A \sim 185$–190) there also exist pancake-shaped or oblate nuclei. Whether there also exist triaxial nuclei is still open to question (see Secs. 1.5.3 and 3.3.3).

(iii) **Strongly enhanced quadrupole transition probabilities.** In the rotational model, the quadrupole transition probabilities are directly connected to the intrinsic quadrupole moment Q_0 [see Eq. (1.73)]. The strongly enhanced BE2-values within the rotational bands are, therefore, another indication of stable quadrupole deformations.

(iv) **Hexadecupole matrix elements.** In (α, α') scattering [HGH 68] and Coulomb excitation, extremely large hexadecupole matrix elements have been found. This is a hint about stable hexadecupole deformations β_4. They are positive (diamond shaped figures; see Fig. 1.3) at the beginning, go to zero in the middle, and turn slightly negative at the end of the rare earth region.

(v) **Single particle structure.** A very sensitive experimental test for the deformation comes about from the experimentally observed single-particle energies, which depend very much on the details of the deformation (see Fig. 2.21).

(vi) **Fission isomers.** In some of the heaviest nuclei are found long

lived states (shape isomers $\tau \lesssim 1$ msec) that have extremely large deformations ($\beta \sim 0.6$) [PDK 62, St 66, Va 77]. The corresponding rotational spectra have been measured [SWK 72, Sp 74] and we can deduce the corresponding moments of inertia. It is clear that such large deformations play an important role in nuclear fission, and the deformed shell model will therefore also be of importance for this process.

Microscopic Hartree–Fock calculations in this region result in a deformed average potential and are indeed able to explain quantitatively the experimental findings (see Chap. 7). Conceptually, the deformed shell model is more involved than the spherical one, since in the formulation of the former we have to accept another symmetry violation—that of rotational invariance.

2.8.2 General Deformed Potential

If we accept the arguments given in Section 2.8.1 about stable nuclear deformation, we are naturally led to the assumption that the average nuclear potential is also deformed. Since the nuclear forces have a small range (~ 1 fm) compared to the nuclear diameter, one expects that the shape of this potential will be similar to the shape of the nuclear density distribution (which can be determined at least in principle from experimental data, e.g., the quadrupole or higher multipole moments). As we have already seen in the case of the spherical shell model, the Woods–Saxon potential (Eq. (2.4)) represents quite a good average potential. It is thus natural to generalize it to the deformed case* [FS 66, DPP 69, GPA 71]

$$V(r,\theta,\phi) = -V_0\left[1+\exp\left(\frac{r-R(\theta,\phi)}{a(\theta,\phi)}\right)\right]^{-1}. \tag{2.72}$$

In the spherical case, the parameter a describes the surface diffuseness and is approximately constant over all spherical nuclei, and therefore does not depend on the curvature of the surface. To get such a constant surface diffuseness for deformed nuclei also, one has to allow for a small dependence of $a(\theta,\phi)$ on the angles θ,ϕ (for more details, see [BM 75] and [BDJ 72]).

As we have seen in Section 2.4, the spin orbit term plays a very important role for the explanation of the level structure of spherical nuclei. In the deformed region, we also have to take it into account and a straightforward generalization of Eq. (2.18) is given by

$$V_{LS} = \lambda(\nabla V(r,\theta,\phi) \wedge \mathbf{p})\mathbf{s}. \tag{2.73}$$

This definition coincides with the one for the spherical case.

*For single-particle energies and wave functions of rare earth nuclei in a deformed Woods–Saxon potential, see [GIS 73].

As in the case of the spherical Woods–Saxon potential, another useful approximation to the general potential (2.72) and (2.73) is the harmonic oscillator, which has then, of course, to be deformed. This study, which we shall discuss now, was orginally carried out by S. G. Nilsson [Ni 55].

2.8.3 The Anisotropic Harmonic Oscillator

If we suppose for the moment that the density of a deformed nucleus can be ideally represented by an ellipsoidal distribution, then it follows from what has been said above that the average potential should also be ellipsoidal. In the harmonic oscillator approximation to the potential (2.72), this is most easily achieved by using the anisotropic harmonic oscillator as average field:

$$h_0 = -\frac{\hbar^2}{2m}\Delta + \frac{m}{2}\left(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2\right). \tag{2.74}$$

The three frequencies $\omega_x, \omega_y, \omega_z$ have to be chosen proportional to the inverse of the half axes a_x, a_y, a_z of the ellipsoid:

$$\omega_\nu = \mathring{\omega}_0 \frac{R_0}{a_\nu}, \qquad (\nu = x, y, z). \tag{2.75}$$

The condition for volume conservation is, therefore,

$$\omega_x \omega_y \omega_z = \text{const.} = \mathring{\omega}_0^3. \tag{2.76}$$

The Hamiltonian (2.74) is separable in x, y, z. The eigenstates are characterized by the quantum numbers n_x, n_y, n_z, and the eigenvalues are:

$$\epsilon_0(n_x, n_y, n_z) = \hbar\omega_x\left(n_x + \tfrac{1}{2}\right) + \hbar\omega_y\left(n_y + \tfrac{1}{2}\right) + \hbar\omega_z\left(n_z + \tfrac{1}{2}\right). \tag{2.77}$$

In the case of *axially symmetric shapes*, one usually chooses the z-axis as symmetry axis and introduces a deformation parameter δ by the following definition:

$$\begin{aligned} \omega_\perp^2 = \omega_x^2 = \omega_y^2 &=: \omega_0^2(\delta)\left(1 + \tfrac{2}{3}\delta\right), \\ \omega_z^2 &=: \omega_0^2(\delta)\left(1 - \tfrac{4}{3}\delta\right), \end{aligned} \tag{2.78}$$

where δ is the only deformation parameter in this case, since $\omega_0(\delta)$ is determined in such a way that volume conservation is guaranteed. Up to quadratic terms in δ, we get from Eq. (2.76),

$$\omega_0(\delta) = \mathring{\omega}_0\left(1 + \tfrac{2}{3}\delta^2\right). \tag{2.79}$$

Nilsson therefore introduced a deformation-dependent oscillator length $b(\delta) = (\hbar/m\omega_0(\delta))^{1/2}$ and dimensionless coordinates $\mathbf{r}' = \mathbf{r}/b$. In these coordinates the Hamiltonian (2.74) has the form

$$h_0(\delta) = \hbar\omega_0(\delta)\left(-\frac{1}{2}\Delta' + \frac{1}{2}r'^2 - \frac{1}{3}\sqrt{\frac{16\pi}{5}}\,\delta r'^2 Y_{20}(\theta', \phi')\right). \tag{2.80}$$

The equipotential surfaces are ellipsoids. In first order in the deformation δ they can be represented by Eq. (1.7).

$$r' \sim (1 + \beta Y_{20}(\theta', \phi')), \tag{2.81}$$

with

$$\beta = \frac{1}{3}\sqrt{\frac{16\pi}{5}}\ \delta + \cdots = 1.057\delta + \cdots. \tag{2.82}$$

The deformation parameter δ of Eq. (2.78) is therefore roughly equal to β [Eq. (1.13)].

In the case of axial symmetry, it is convenient to use cylinder coordinates [Fl 71]. The eigenstates are characterized by quantum numbers n_z, n_ρ, m_l, where m_l is the projection of the orbital angular momentum on to the symmetry axis. With

$$N = n_z + 2n_\rho + m_l = n_x + n_y + n_z, \tag{2.83}$$

we get from Eq. (2.77)

$$\epsilon_0(n_z, n_\rho, m_l) = \hbar\omega_z\left(n_z + \tfrac{1}{2}\right) + \hbar\omega_\perp(2n_\rho + m_l + 1) \tag{2.84}$$

$$\simeq \hbar\mathring{\omega}_0\left\{\left(N + \frac{3}{2}\right) + \delta\left(\frac{N}{3} - n_z\right)\right\}. \tag{2.85}$$

The axial symmetry causes m_l to be a good quantum number. The same is true for the spin component s_z and the z-component j_z of the total angular momentum, which has the eigenvalue

$$\Omega = m_l + m_s = m_l \pm \tfrac{1}{2}. \tag{2.86}$$

It is usual to characterize the eigenstates of h_0 in the cylindrical basis by the set of "Nilsson" quantum numbers

$$\Omega\pi[Nn_z m_l], \tag{2.87}$$

where π is the parity of the states [$\pi = (-1)^l = (-1)^N$; see Eq. (2.9)].

To discuss the level structure for a definite example, we use $N = 3$. From Eq. (2.85), we obtain in this case

$$\epsilon_0^{N=3}(n_z n_\rho m_l) \simeq \frac{9}{2}\hbar\mathring{\omega}_0 + \hbar\mathring{\omega}_0\delta(1 - n_z). \tag{2.88}$$

Table 2.2 Construction of the Nilsson quantum numbers for $N = 3$

n_z	m_l	n_ρ	Ω	degeneracy
0	1 3	1 0	1/2 3/2 5/2 7/2	4-fold
1	0 2	1 0	1/2 3/2 5/2	3-fold
2	1	0	1/2 3/2	2-fold
3	0	0	1/2	1-fold

Using Eqs. (2.83) and (2.86), we get the following possibilities for the different quantum numbers, which are displayed in Table 2.2.

According to Eqs. (2.85) and (2.88), the levels with different values of n_z are split for small deformations proportionally to δ. This is shown in Fig. 2.19.

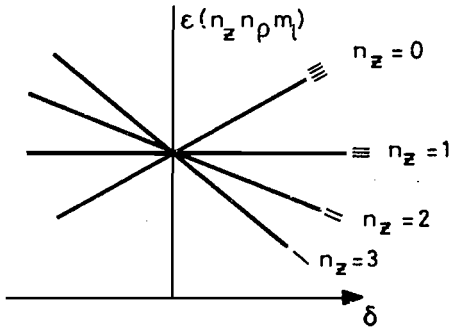


Figure 2.19. Levels of the anisotropic harmonic oscillator as a function of δ.

2.8.4 Nilsson Hamiltonian

As we have seen in Sec. 2.3, the pure harmonic oscillator has two essential drawbacks concerning the agreement with experimental single-particle spectra:

(i) A strong spin orbit term must be added in order to reproduce the right magic numbers
(ii) For heavy nuclei, the realistic average potential is rather flat in the interior of the nucleus. Compared to the harmonic oscillator, nucleons at the surface (i.e., nucleons with higher l-values) feel a deeper potential in the realistic case.

In order to include these effects, Nilsson [Ni 55] added two terms to the deformed harmonic oscillator (2.74) and (2.80) and used the Hamiltonian:

$$h = -\frac{\hbar^2}{2m}\Delta + \frac{m}{2}\omega_\perp^2(x^2+y^2) + \frac{m}{2}\omega_z^2 z^2 + C\mathbf{ls} + D\mathbf{l}^2 \tag{2.89}$$

$$= \hbar\omega_0(\delta)\left(-\frac{1}{2}\Delta' + \frac{1}{2}r'^2 - \beta r'^2 Y_{20}\right) - \kappa\hbar\dot{\omega}_0(2\mathbf{ls} + \mu\mathbf{l}^2). \tag{2.90}$$

The constants C and D are given in the form:

$$C = -2\hbar\dot{\omega}_0\kappa, \qquad D = -\hbar\dot{\omega}_0\kappa\mu, \tag{2.91}$$

where C gives the strength of the spin orbit force and $D\cdot\mathbf{l}^2$ shifts the levels with higher l-values downward (Fig. 2.20; notice that different μ-values are taken for different shells, as explained below).

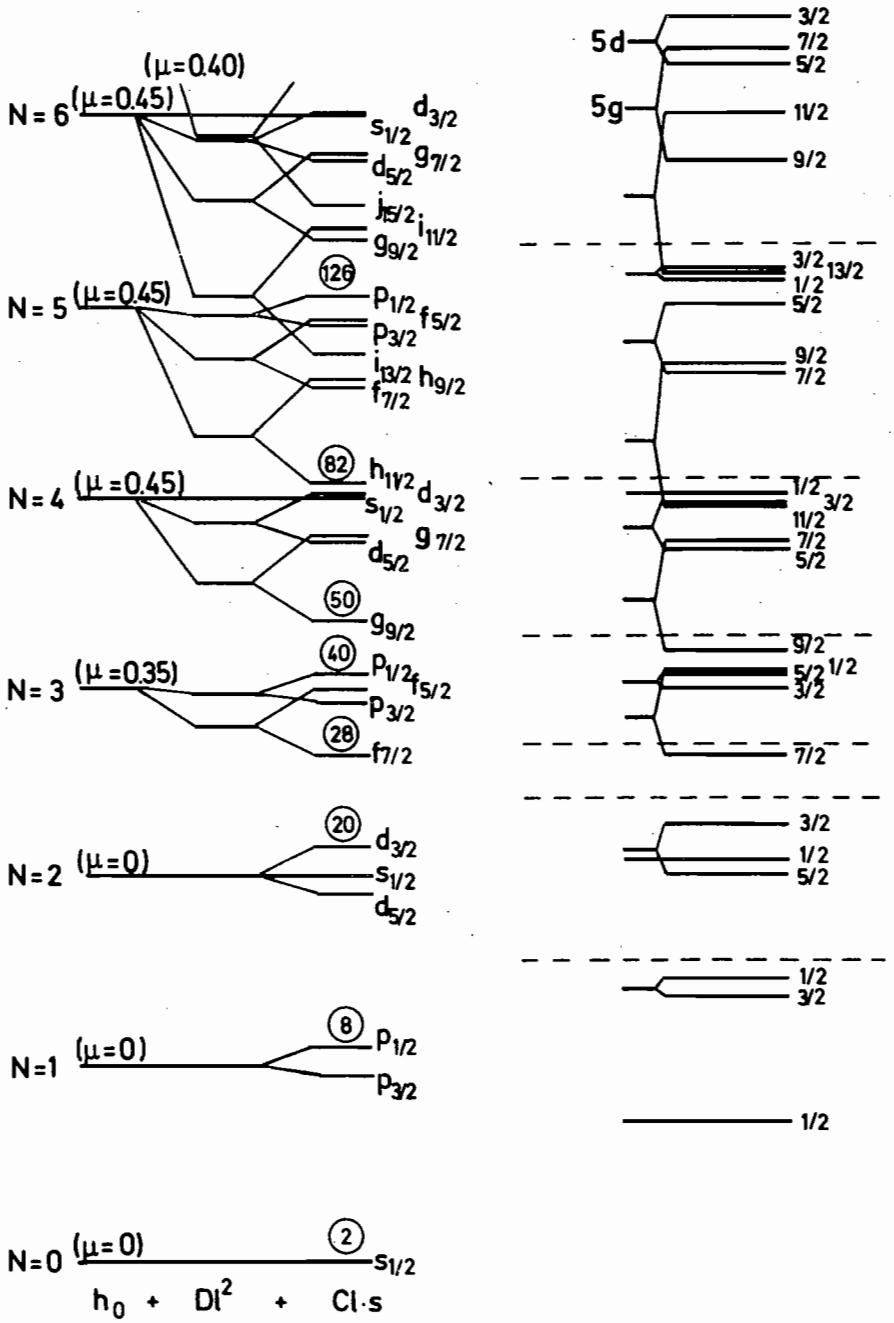


Figure 2.20. Comparison of experimentally determined level scheme [Kl 52] with calculations [Ni 55] using the Nilsson Hamiltonian for zero deformation.

In the original version, Nilsson used the term $D \cdot \mathbf{l}^2$. Later on one observed that for states with large N quantum numbers, the corresponding shift is too strong and the following replacement was made [GLN 67]

$$D \cdot (\mathbf{l}^2 - \langle \mathbf{l}^2 \rangle_N), \tag{2.92}$$

where $\langle \mathbf{l}^2 \rangle_N = \frac{1}{2} \cdot N(N+3)$ is the expectation value of $\mathbf{l}^2$ averaged over one major shell with quantum number N. In this case, only the states within the shells are shifted. The center of gravity between different major shells remains unaffected.

Of course, the $\mathbf{l} \cdot \mathbf{s}$ and the $\mathbf{l}^2$ term are no longer diagonal in the representation $n_z\, n_\rho\, m_l$, nor in the representation n_x, n_y, n_z. The only quantum numbers that remain conserved are the parity π and the eigenvalue Ω of j_z. In fact, it is easily shown that h of Eq. (2.90) does not commute with $\mathbf{j}^2$; therefore the Slater determinant for a deformed shell model has no good quantum number for the total angular momentum. This means that the picture of a deformed nucleus, that is, a deformed shell model, is inevitably linked to the abandonment of rotational invariance. In nature, however, rotational invariance is never violated and one should therefore, at least in principle, re-establish this invariance before drawing any conclusions. How this can be done will be shown in Sec. 11.4. Approximate treatments of these so-called projection methods show that the main features of the deformed shell model are correct.

For large deformations, the $\mathbf{ls}$ and $\mathbf{l}^2$ terms in Eq. (2.90) can be neglected in comparison with βY_{20}. In this limit, the quantum numbers (2.87) of the anisotropic harmonic oscillator become good quantum numbers. They are then also termed *asymptotic quantum numbers*.

In order to obtain the eigenvalues of the Nilsson Hamiltonian as a function of δ, it must be diagonalized in a suitable basis. The isotropic or anisotropic harmonic oscillator [BP 71] can be used as a basis set. In his original paper, Nilsson [Ni 55] worked in a spherical basis. In this case, the $\mathbf{l}^2$ and $\mathbf{ls}$ terms are diagonal and only the term βY_{20} mixes states with the same principal quantum number N ($\Delta N = 0$) with those with $\Delta N = 2$. By using stretched coordinates Nilsson could show [Ni 55] that the $\Delta N = 2$ admixtures are only of higher order in the deformation parameter and can therefore be neglected to a good approximation. The deformation parameters in the stretched coordinates are usually called ϵ_i. Because of the $\mathbf{ls}$ and $\mathbf{l}^2$ terms, the levels with the same n_z in Fig. 2.19 split up and only the degeneracy $\pm\Omega$ is conserved. Nevertheless, the levels with the same n_z are, in the asymptotic region, nearly parallel. This effect can be seen in the so-called Nilsson diagram of Fig. 2.21a, for instance, for the levels with $N = 3$. In Figs. 2.21b and 2.21c the Nilsson diagrams for higher shells are given for neutrons and protons, respectively. Again, we can see that levels with the same n_z are parallel in the asymptotic region. For the correct attribution of the n_z quantum number, one should also notice that according to Eq. (2.85), levels with higher (lower) n_z values are lower than those

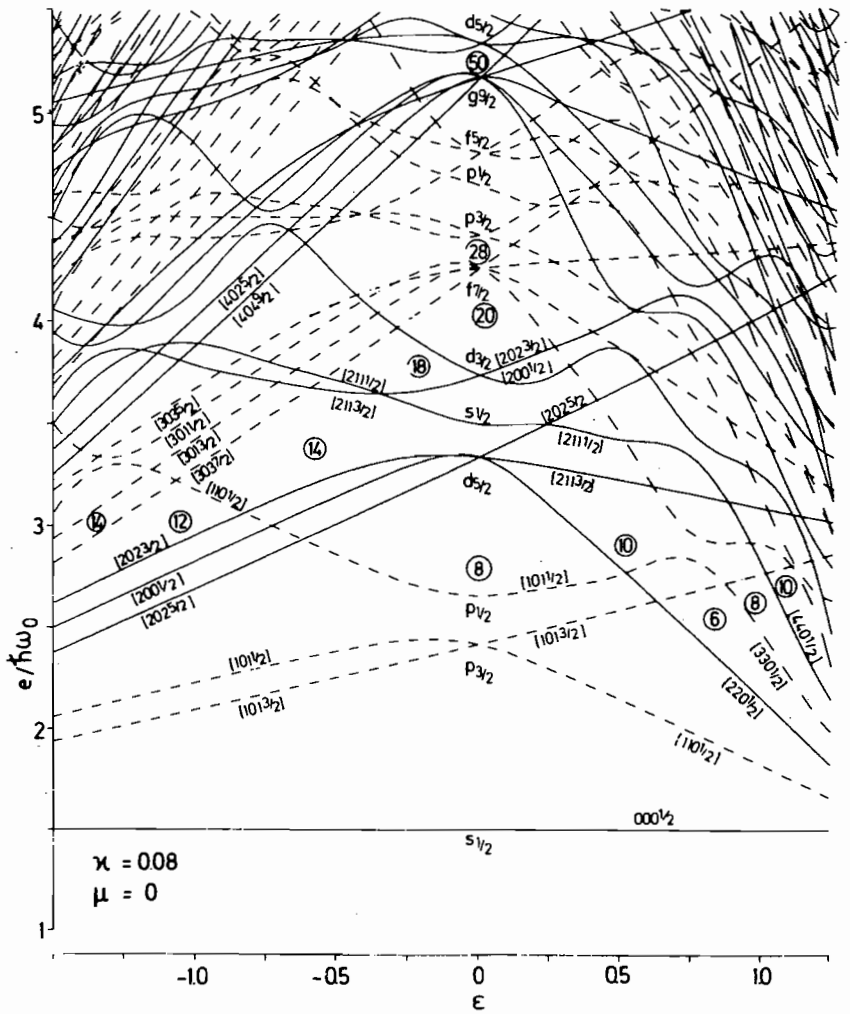


Figure 2.21a. Level scheme of the Nilsson Model for light nuclei. The figure shows the single-particle energies e [the eigenvalues of the Nilsson Hamiltonian (2.90)] in units of the oscillator energy $\hbar\omega_0(\epsilon)$ at the corresponding deformations $\epsilon = \epsilon_2$ (stretched coordinate; see text). The full lines correspond to levels with positive parity, the dashed lines to those with negative parity. The numbers on the lines correspond to the quantum numbers N, n_z, m_l, Ω. Also indicated are the magic numbers in the spherical case and for finite deformations. (We are grateful to Dr. S. Åberg for the preparation of the figure.)

with lower (higher) n_z-values for positive (negative) δ. Knowing N, n_z, and Ω one can also determine the m_l value, and hence the complete set of asymptotic quantum numbers, from tables analogous to Table 2.2.

The Nilsson Hamiltonian (2.90) contains no Coulomb term. The effect of that term is incorporated into an appropriate choice of the constants κ and μ. They are actually fitted such that the observed levels in deformed nuclei are reproduced. The spherical single-particle energies that one obtains with these parameters agree qualitatively with the single-particle or

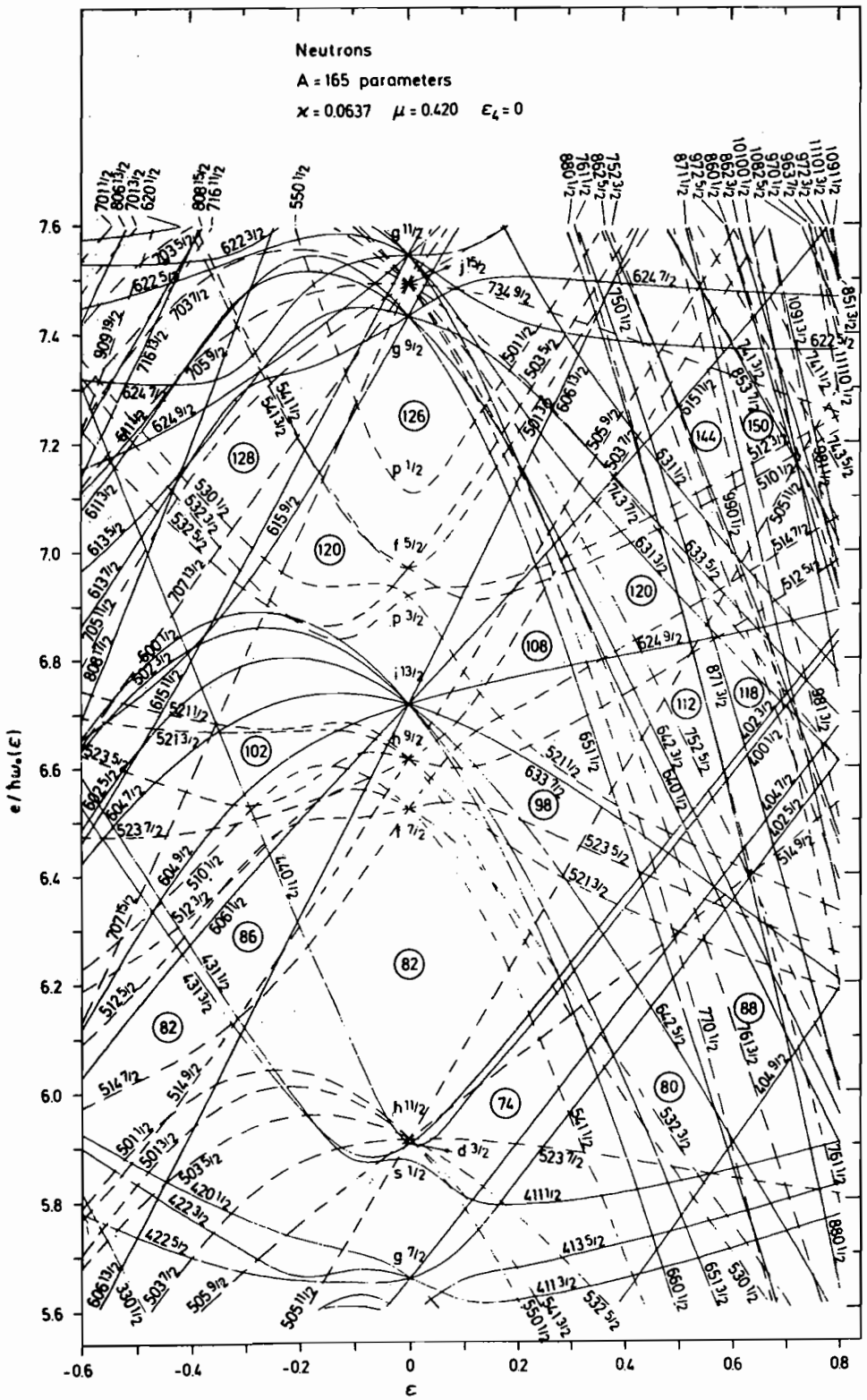


Figure 2.21b. Same as Fig. 2.21a for neutrons in heavy nuclei.

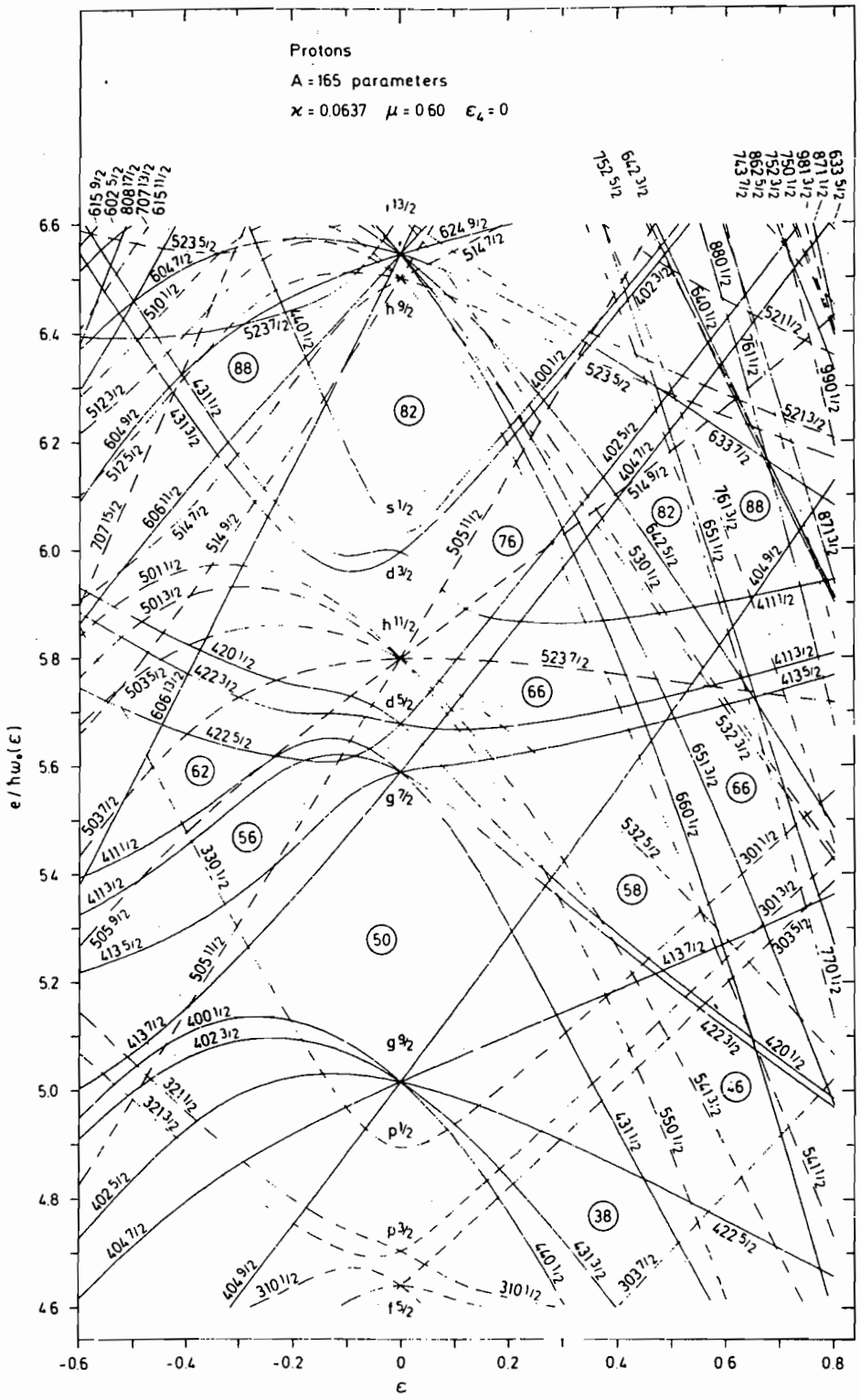


Figure 2.21c. Same as Fig. 2.21a for protons in heavy nuclei.

single-hole spectra of near magic nuclei. However, sometimes there are quantitative discrepancies. For a discussion of this point, see [RL 76].

To get a good fit, different values of κ and μ for different shells are used. In his original paper, Nilsson used $\kappa=0.05$ for all shells and $\mu=0$ for $N=0,1,2$; $\mu=0.35$ for $N=3$; $\mu=0.45$ for $n=4,5,6$; and $\mu=0.4$ for $N=7$. Later these parameters were more carefully adjusted. Table 2.3 gives values that are now widely used. There the same values of μ and κ are used for all the shells, but they depend on the nucleus (N, Z) one is interested in.

Table 2.3 Parameters of the Nilsson Hamiltonian (from [GLN 67])

Region	κ	μ
$N, Z<50$	0.08	0
$50<Z<82$	0.0637	0.60
$82<N<126$	0.0637	0.42
$82<Z$	0.0577	0.65
$126<N$	0.0635	0.325

There are many characteristic features in the Nilsson diagram, some of which we want to mention here (however, for a more complete discussion, we refer the reader to Nilsson's original paper [Ni 55]):

(i) The shells which are determined by the single particle angular momentum j at zero deformation, split up into $(2j+1)/2$ levels for $\delta\neq 0$. Each of these is twofold degenerate with eigenvalues $\pm\Omega$ ($[h, j_z]=0$) and can therefore be characterized by $|\Omega|$ and its parity. The quantum numbers $[Nn_z m_l]$ are not conserved for small deformations, nevertheless they are used to classify the levels.

(ii) The quadrupole field r'^2Y_{20} causes the levels with lower Ω values to be shifted downwards for positive deformations (prolate shapes) and to be shifted upwards for negative deformations (oblate shapes). One can understand this effect, realizing that the states with low Ω-values have a relatively higher probability of being close to the z-axis. This corresponds to a positive quadrupole moment $\langle r'^2Y_{20}\rangle$. Because of the minus sign in Eq. (2.90), their energy is shifted downwards. The nucleons with a prolate density distribution $|\phi_i(\mathbf{r})|^2$ lie deeper in the deformed well.

(iii) For larger deformations, it can happen that the levels change their slope, for example, there are two $1/2$-states in the $N=1$ shell. This results from the interaction of two levels with the same quantum numbers $\Omega\pi$ coming from different j-shells. As a rule, levels of this kind can never cross (Neumann–Wigner no crossing rule; Fig. 2.22) [NW 29, LL 59, Vol. 3, Chap. 11; HW 53].

The repulsion $\Delta\epsilon$ at the crossing point is proportional to the interaction strength. Properties of the levels become interchanged at

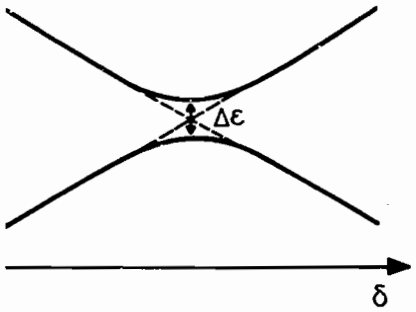


Figure 2.22. No crossing rule for two levels with the same symmetry.

the crossing point and the wave functions corresponding to the two levels far from the crossing point are the same as if there had been no interaction at all.

(iv) If one diagonalizes the Nilsson Hamiltonian using the basis $|Nlj\Omega\rangle$ of the isotropic harmonic oscillator, the Nilsson wave functions are given by a superposition of spherical harmonic oscillator functions:

$$|k\rangle = \sum_{\alpha} D_{\alpha k}|\alpha\rangle, \tag{2.93}$$

where

$$\alpha = \{Nlj\Omega\}.$$

For small values of δ, there is only small mixing, that is, one of the coefficients $D_{\alpha k}$ is nearly equal to one, and the others are close to zero. For larger deformations, the situation depends very much on the specific levels. For example, the $1g_{9/2}$ state in Fig. 2.21c splits up into five levels $\Omega = 1/2, \ldots, 9/2$. Since for $\delta = 0$ the $g_{9/2}$ is rather far away from the other levels with the same N (it is lowered by the spin orbit term), there is only a small amount of mixing, and the corresponding levels are almost eigenstates of $\mathbf{j}^2$, even for quite large values of δ. The same effect is even more pronounced for heavier nuclei. For instance, the $i_{13/2}$ state is of this type and plays an important role in the rare earth region and contains almost no mixing for relevant deformation values.

(v) The slope of the Nilsson levels ϵ_k [measured in units of $\hbar\omega_0(\epsilon)$] is given by the single-particle matrix element of the quadrupole operator $q = r'^2 Y_{20}$ in the corresponding single-particle state $|k\rangle$

$$\frac{d\epsilon_k}{d\beta} = -\langle k|r'^2 Y_{20}|k\rangle.$$

To prove this relation, we use the fact that the eigenvalues

$$\epsilon_k = \langle k|h(\beta = 0) - \beta q|k\rangle$$

of the diagonalization problem (2.90) are stationary with respect to small variations of the functions $|k\rangle$ (see Sect. 5.2), that is,

$$\delta\langle k|h(\beta = 0)|k\rangle - \beta\,\delta\langle k|q|k\rangle = 0.$$

We thus obtain

$$\frac{d\epsilon_k}{d\beta} = \frac{d}{d\beta}\langle k|h(\beta=0)|k\rangle - \beta\frac{d}{d\beta}\langle k|q|k\rangle - \langle k|q|k\rangle = 0 - \langle k|q|k\rangle.$$

2.8.5 Quantum Numbers of the Ground State in Odd Nuclei

The spherical shell model fails in its prediction for the angular momenta and parities of the ground state for nuclei in the deformed region. However, the Nilsson model is able to explain the experimental data in the following way [MN 59].

For a specific nucleus, one determines the deformation parameter δ from the measured quadrupole moment Q_0, as discussed in Section 2.8.1. For this value of δ one successively fills two protons and two neutrons in each level starting from the bottom. Because of the degeneracy of each Nilsson level ($\pm\Omega$), every nucleon pair has, qualitatively speaking, spin zero. Therefore spin and parity of odd nucleons are determined by the last odd nucleon. Since Ω is not a directly measurable quantity, we must determine it indirectly by using a model. We will see in Section 3.3.1.1 that the appropriate concept here is the particle-plus-rotor model. There it will turn out that the angular momentum I_0 of the band head of a rotational band coincides with Ω ($I_0=\Omega$). The lowest band head is, therefore, the ground state spin of the odd nucleus. Often in the region of the experimentally determined value of δ several possibilities for the Ω of the last particle exist. In fact, the observed value of the ground state spin for many deformed nuclei is among the possible Ω values indicated in Table 2.4. In Fig. 2.23 this method is explained for $^{183}_{74}W_{109}$. The other possible Ω values

Table 2.4 Comparison of theoretically and experimentally determined ground state spins [He 61, p. 648]

	δ	$I_{0\,\text{theo}}$	$I_{0\,\text{exp}}$		δ	$I_{0\,\text{theo}}$	$I_{0\,\text{exp}}$
$_{63}Eu^{151}$	0.16	$3/2^{\pm}, 5/2^{\pm}, 1/2^-$	5/2	Gd^{155}_{91}	0.31	$5/2^+, 3/2^-$	3/2
$_{63}Eu^{153}$	0.30	$5/2^+, 3/2^+$	5/2	Gd^{157}_{93}	0.31	$3/2^-, 5/2^+$	3/2
$_{65}Tb^{159}$	0.31	$3/2^+, 5/2^+$	3/2	Dy^{161}_{95}	0.31	$5/2^-$	
$_{67}Ho^{165}$	0.30	$7/2^-, 1/2^+$	7/2	Er^{167}_{99}	0.29	$1/2^-, 7/2^+, 11/2^-$	7/2
$_{69}Tm^{169}$	0.28	$1/2^+, 7/2^-$	1/2	Yb^{171}_{101}	0.29	$7/2^+, 1/2^-, 11/2^-$	1/2
$_{71}Lu^{175}$	0.28	$7/2^+, 5/2^+$	7/2	Yb^{173}_{103}	0.29	$5/2^-$	5/2
$_{73}Ta^{181}$	0.23	$5/2^+, 7/2^+$	7/2	Hf^{177}_{105}	0.26	$7/2^-$	7/2
$_{75}Re^{183}$	0.19	$9/2^-, (5/2^+)$	5/2	Hf^{179}_{107}	0.27	$9/2^+$	9/2
$_{75}Re^{187}$	0.19	$9/2^-, (5/2^+)$	5/2	W^{183}_{109}	0.21	$1/2^-, 7/2^-, 3/2^-$	1/2
$_{77}Ir^{191}$	0.14	$3/2^+, 1/2^+, 11/2^-$	3/2	Os^{187}_{111}	0.18	$1/2^-, 3/2^-, 9/2^-$	1/2
$_{77}Ir^{193}$	0.12	$3/2^+, 1/2^+, 11/2^-$	3/2	Os^{189}_{113}	0.15	$1/2^-, 3/2^-, 11/2^+, (9/2^-)$	3/2

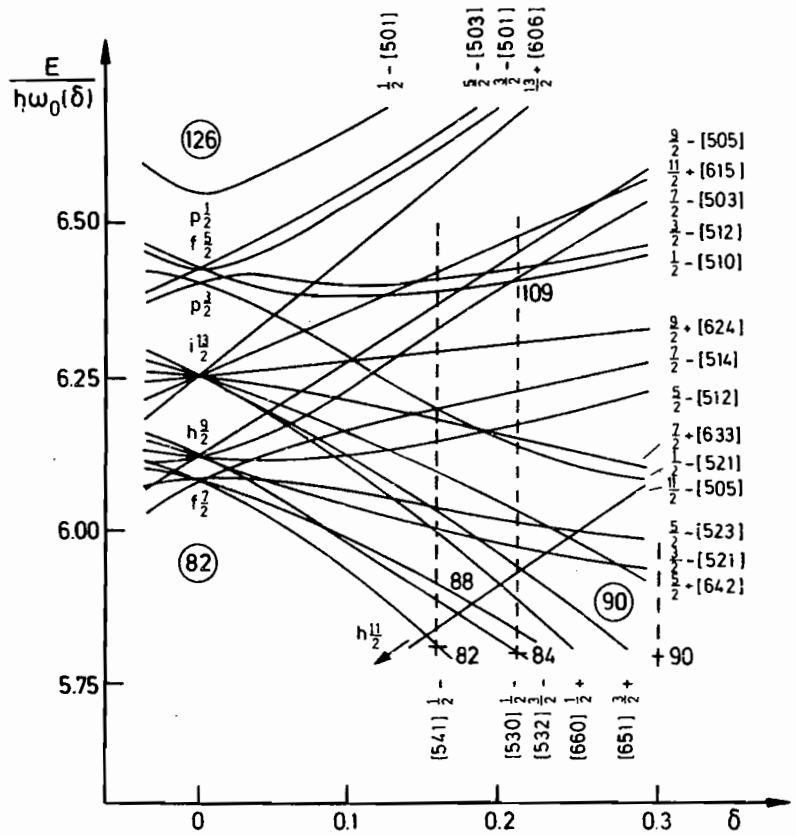


Figure 2.23. Determination of the ground state spin in the Nilsson model. (From[He 61, p. 647].)

are usually seen in the spectra as band heads of excited rotational bands, which is a nice success of the Nilsson model. Single-particle energies in this scheme can be found in [Ch 66, BR 71a, OWP 71, GIS 73, GSK 75, CAF 77].

2.8.6 Calculation of Deformation Energies

In order to obtain the ground state energy of the nucleus in the (spherical or deformed) shell model, we cannot simply sum the single-particle energies from the bottom of the potential up to the Fermi level, because the effect of the nucleon–nucleon interaction would be counted twice (see also Sec. 5.3). To account for this, we proceed with the following qualitative considerations.

We assume that the average potential which is felt by the ith nucleon is given by the sum of all two-body potentials acting on this particle:

$$V_i = \sum_{j \neq i} v_{ij}. \tag{2.94}$$

The Hamilton operator

$$H=\sum_{i=1}^{A} t_i+\tfrac{1}{2}\sum_{\substack{i,\,j\\ i\neq j}} v_{ij} \tag{2.95}$$

is then given by

$$H=\tfrac{1}{2}\sum_{i=1}^{A} h_i+\tfrac{1}{2}\sum_{i=1}^{A} t_i; \qquad h_i=t_i+V_i. \tag{2.96}$$

If V_i has the form of a harmonic oscillator, we have, because of the virial theorem [Da 65a, Chap. II],

$$\langle t_i\rangle=\langle V_i\rangle=\tfrac{1}{2}\langle h_i\rangle \tag{2.97}$$

and we get as the ground state energy

$$E_0(\delta)=\langle H\rangle=\tfrac{3}{4}\sum_{i=1}^{A}\langle h_i\rangle=\tfrac{3}{4}\sum_{i=1}^{A}\epsilon_i(\delta). \tag{2.98}$$

In practice we determine the equilibrium deformation δ_{eq} in calculating E_0 as a function of δ; the absolute minimum then determines δ_{eq}. This procedure does not, of course, include any residual interaction and, in fact, fails to reproduce the absolute value of the binding energies. Only the resulting equilibrium deformations are approximately right (see Fig. 2.24). Further investigations on these lines [BS 61, Sz 61, So 67, and GLN 67] have taken into account the following additional points.

(i) A *residual interaction* of the pairing type (see Chap. 6)
(ii) The expectation value of the *Coulomb force*

$$E_{\text{Coul}}=\Big\langle \sum_{\text{prot}} \frac{e^2}{|\mathbf{r}_i-\mathbf{r}_j|}-\kappa\hbar\mathring{\omega}_0\Delta\mu\sum_{\text{prot}}\left(\mathbf{l}^2-\langle\mathbf{l}^2\rangle_N\right)\Big\rangle, \tag{2.99}$$

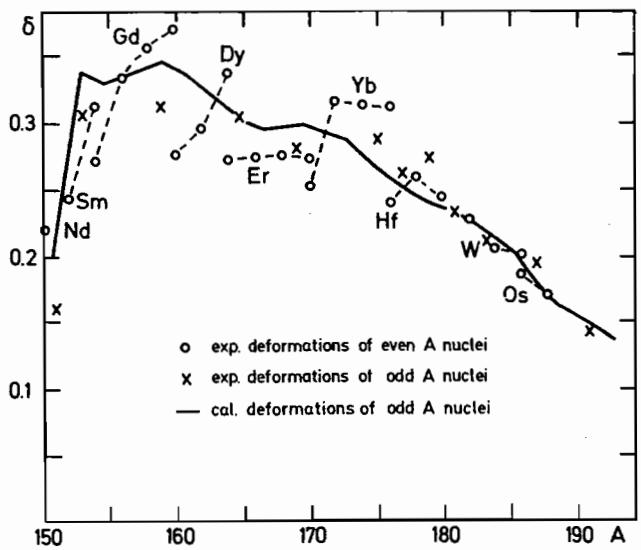


Figure 2.24. Comparison of theoretically and experimentally determined deformations. (From [MN 55].)

where $\Delta\mu = \mu_{\text{prot}} - \mu_{\text{neut}}$ corrects for the Coulomb effect already taken into account by the difference between the proton and neutron parameters μ. It turns out that the effect of the Coulomb repulsion favors larger deformations, whereas the effect of the pairing correlations is the converse. On the whole, the results of the old Nilsson model are reproduced, because Coulomb and pairing forces counteract one another at the equilibrium point.

(iii) *Higher shapes.* It is evident that quadrupole deformations alone are not able to describe for instance the fission process. On the other hand, there exists evidence for hexadecupole deformations in the ground state of some nuclei [HGH 68]. It is possible to describe such effects within the Nilsson model by including, besides Y_{20}, higher terms such as Y_{40} and Y_{60} [NTS 69, Mö 72, MN 73, RNS 78].

For the fission process it is often useful to introduce deformed potentials that allow us to describe two separated fragments at large distances. In contrast to the Nilsson model, such potentials have the right asymptotic behavior. Several versions of such potentials have been used:

(i) *Two-center models* [DR 66, HMG 69, ADD 70, GMG 71, SGM 71, MG 72, MMS 73], which are based on two oscillators with separated centers.

(ii) The *Folded Yukawa Potential* [Ni 69, BFN 72, Ni 72, MN 73] which starts with a density of sharp surface but arbitrary shape. To get the corresponding potential, it is folded with a short-range Yukawa force.

(iii) *Generalized Wood–Saxon* potentials with a deformation-dependent surface thickness [DPP 69, BDJ 72, Pa 73, BLP 74, JH 77].

The method of simply summing up the single-particle energies fails to reproduce the absolute binding energies and to describe the energy surface at very large distortions. The reason for this is that the binding energy is a bulk property. In fact, rather small shifts in the single-particle energies produce large errors in the binding energy. To get the proper values for the binding energy together with shell effects, we must use a combination of the liquid drop and the shell model, as proposed by Strutinski (see Sec. 2.9). Calculations within this method show that the pure shell model as described in this chapter allows us to determine only ground state deformations.

The fact that the absolute energy minimum occurs at finite values of δ for nuclei between closed shells, can be understood qualitatively by considering the level density as a function of deformation. It turns out that for quite general average potentials the level density develops shell effects for certain definite values of the deformation; that is, at these deformations the levels are not randomly distributed as a first glance on the Nilsson

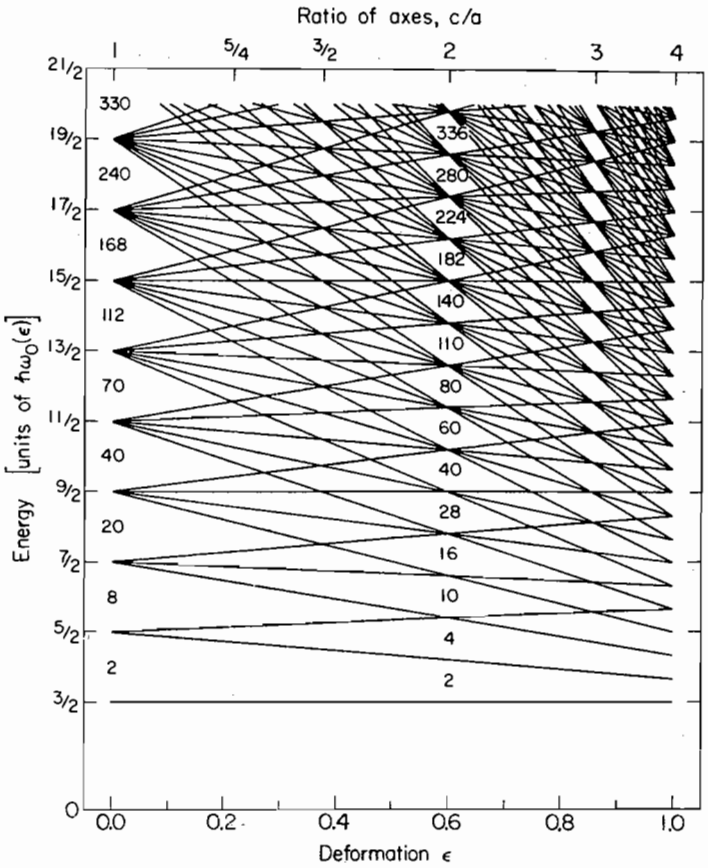


Figure 2.25. Energy levels of an harmonic-oscillator potential for prolate spheroidal deformations ϵ. (From [MN 73].)

diagram would indicate, but are, for instance, grouped in bunches at larger deformations.* This is shown in Fig. (2.25) for the harmonic oscillator.

If one plots the energy as a function of the deformation, one finds that whenever the Fermi level is situated in a low density region, the nucleus is more bound, whereas it is less bound when the Fermi level is in a high density region. This, in turn, means that the occupied levels are, on the average, more bound if the Fermi level is in a low density region than if it is in a high density region. Since for nuclei between closed shells the Fermi level for $\delta = 0$ is in a region of high level density, it becomes qualitatively understandable that these nuclei want to deform into a region where the Fermi energy is in a lower level density region. From this it also becomes clear that in any nucleus more than one minimum, as a function of deformation, can eventually develop. As we discussed in Section 2.8.2, in very heavy nuclei we find second minima at about $\delta \simeq 0.6$, somewhat

*One can understand from semiclassical arguments, at which deformations such shell closures have to occur [BM 75, St 75b, SM 76, SMO 77].

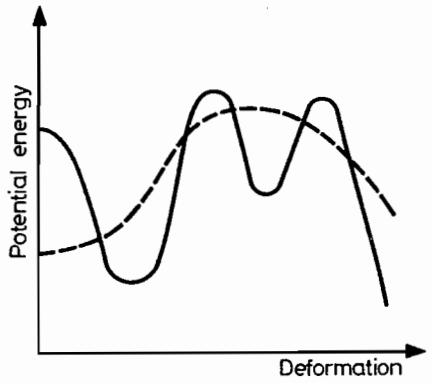


Figure 2.26. Schematic variation of the energy with deformation for a nucleus with a second minimum. The dashed line corresponds to the liquid drop barrier.

higher than the first minimum, which gives rise to the so-called shape isomeric states [St 66]. Qualitatively, the energy as a function of deformation is shown in Fig. 2.26.

2.9 Shell Corrections to the Liquid Drop Model and the Strutinski Method

2.9.1 Introduction

Up to now we have studied two quite different descriptions of the atomic nucleus. The *liquid drop model* (LDM) assumes that the nucleons produce a spatially uniform density distribution in the nucleus with a sharp edge at the surface. It is able to reproduce the overall features of the nucleus, that is, most properties that depend only in a smooth way on the nucleon number, as, for example, in Chapter 1, the A dependence of the nuclear binding energy (Fig. 1.2). On the other hand, there is the *shell model*. It assumes a quantized independent particle motion in an average potential to be valid, and we have seen in Section 2.3 and 2.8 that this model reproduces nicely those particular nuclear properties in which only the nucleons in the vicinity of the Fermi surface are involved.

Phenomenological shell models (in contrast to Hartree–Fock calculations; see Chap. 5), however, fail to correctly reproduce properties of the nucleus in which *all* nucleons contribute (the so-called bulk properties), like, for instance, the total binding energy. Strutinski [St 67, 68] invented a very elegant method* to reconcile both phenomenological descriptions of the nucleus which eliminates their defects but keeps their qualities. This is the *Strutinski shell correction procedure*. It is able to reproduce not only the

* Investigations on a similar line have been carried out by Myers and Swiatecki [MS 66].

experimental ground state energies of nuclei, but also their dependence on deformation parameters. This method was used extensively for the calculation of energy surfaces in the fission process.

2.9.2 Basic Ideas of the Strutinski Averaging Method

As we discussed in Chapter 1, the nuclear binding energies E as a function of A have a smooth part E_{LDM} well represented by the Bethe–Weizäcker mass formula (1.4) (Fig. 1.2) and, in addition, an oscillatory part E_{osc} defined by

$$E = E_{\mathrm{osc}} + E_{\mathrm{LDM}}. \tag{2.100}$$

Similar oscillations would occur if we could calculate the exact energy of the many-body system as a function of the deformation and compare it with the corresponding LDM value. These oscillations are due to the occurrence of shell closures, that is, they have maxima at the magic numbers, as we mentioned at the beginning of Section 2.2 (Fig. 2.2). Therefore, their origin is entirely of a quantum mechanical nature. As a matter of fact, if we calculate the binding energies in the shell model [for example, in the Nilsson model; see Eq. (2.90)], one finds such oscillations. Only the corresponding average part is wrong. Furthermore, as we explained at the end of the last chapter, these oscillations are due to a grouping of levels into bunches—the shells. In fact, it is clear that for a smooth distribution of levels, the binding energy per particle depends in a smooth way on the position of the Fermi level, whereas for a shell-like distribution of the levels there is also an oscillatory behavior superimposed on top of it. We demonstrate this in Fig. 2.27, where we compare two such level densities whose average densities are equal.

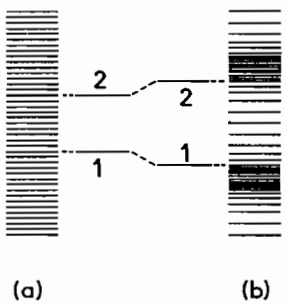


Figure 2.27. Comparison of an equally spaced level density to a schematic shell model level density. For the Fermi level (1), the binding in (b) is stronger than in (a), whereas for (2) the opposite is true.

We see that if the Fermi level is situated just above a shell we get more binding than on the average, whereas if it is placed just below we get less

binding than on the average. This picture is certainly very schematic, since, for example, the real average level density is not a constant; this however, is a smooth effect and should not influence the previous conclusions. The binding energies corresponding to the shell distribution (b) oscillate around the ones given by the "average" level density (a). Therefore, the average level density is responsible for the average behavior of the binding energies. It is this average part which has the wrong value in the phenomenological shell model. It was the decisive idea of Strutinski to calculate only the fluctuating part E_{osc} of the total energy E in Eq. (2.100) within the shell model and to take the rest, E_{LDM}, from the liquid drop model. This procedure contains the assumption that the fluctating part E_{osc} is well approximated by the fluctuating part of the shell model energy. At the end of this chapter we will discuss the way we can justify this assumption. There then remains only the problem of how to divide up the shell model energy into an oscillating part, E_{osc}, and a smoothly varying part, $\tilde{E}_{\text{sh}}$.

$$E_{\text{sh}} = \sum_{i=1}^{A} \epsilon_i = E_{\text{osc}} + \tilde{E}_{\text{sh}}. \tag{2.101}$$

[Expression (2.101) does not take into account any effect coming from the two-body interaction as, for example, in Eq. (2.98). We will see that in the end this is of no importance.]

The decomposition (2.101) is a problem which has to be solved completely within the shell model. Therefore, it is useful to introduce the concept of the level density $g(\epsilon)$ by defining $g(\epsilon)\cdot d\epsilon$ as the number of levels in the energy interval between ϵ and $\epsilon + d\epsilon$. In the shell model the level density is given by

$$g(\epsilon) = \sum_i \delta(\epsilon - \epsilon_i). \tag{2.102}$$

If we know $g(\epsilon)$, we can calculate the particle number

$$A = \int_{-\infty}^{\lambda} g(\epsilon)\, d\epsilon \tag{2.103}$$

with a properly chosen Fermi energy λ.

In the shell model, λ is not defined uniquely by Eq. (2.103). It can be arbitrarily chosen to be between the last filled and the first unfilled level. For the shell model energy, we get

$$E_{\text{sh}} = \int_{-\infty}^{\lambda} \epsilon g(\epsilon)\, d\epsilon. \tag{2.104}$$

The shell model levels are grouped into bunches with an average distance of $\hbar\omega_0 \simeq 41 A^{-1/3}$ (MeV) [see Eq. (2.12)]. Therefore, the level density g shows oscillations with roughly this frequency.

Since, as we have seen in Fig. 2.27, the fluctuations in the shell model energy E_{sh} are due to these oscillations, it is obvious that we can calculate the smooth part $\tilde{E}_{\text{sh}}$ in Eq. (2.101) by introducing a continuous function $\tilde{g}(\epsilon)$, which represents the smooth part of the level density $g(\epsilon)$. It should

have the mean functional behavior of g, but must not contain oscillations with a frequency $\simeq\hbar\omega_0$. In a real nucleus the level density is by far more complex than in our simple picture of Fig. 2.27. Its increase, for instance, with increasing energy is in general nonlinear and certainly far from being a constant. Therefore, one has to give a well-defined mathematical prescription of how one can extract from a given level density its mean part. This will be explained in the next section. For the moment, we shall assume that we are given the average part $\tilde{g}(\epsilon)$ of the shell model level density $g(\epsilon)$. Then we can again calculate a corresponding Fermi energy $\tilde{\lambda}$ by the condition

$$A=\int_{-\infty}^{\tilde{\lambda}}\tilde{g}(\epsilon)\,d\epsilon. \tag{2.105}$$

Since $\tilde{g}$ is continuous, $\tilde{\lambda}$ is well defined by this implicit equation and is usually different from λ. For the smooth part of the energy we finally get

$$\tilde{E}_{\mathrm{sh}}=\int_{-\infty}^{\tilde{\lambda}}\epsilon\tilde{g}(\epsilon)\,d\epsilon. \tag{2.106}$$

The total energy E of the system is therefore given by

$$E=E_{\mathrm{LDM}}+E_{\mathrm{osc}}=E_{\mathrm{LDM}}+E_{\mathrm{sh}}-\tilde{E}_{\mathrm{sh}}. \tag{2.107}$$

As we have seen in Chapter 1, stable liquid drops are always spherical. Because of the additional term E_{osc}, it can happen that in some region of the periodic table the "Strutinski averaged energy" (2.107) has its minimum at finite values of the deformation. However, before we go on to the general discussion of the Strutinski method, let us show how one can define appropriately the average part of the level density.

2.9.3 Determination of the Average Level Density

In this section we have to deal with the problem of how to define in an appropriate mathematical way an average level density, if we are given a shell model density in an infinite three-dimensional well (the restriction to infinite potentials is not essential and the following considerations can be generalized for finite potentials [SI 75]).

$$g(\epsilon)=\sum_{i=1}^{\infty}\delta(\epsilon-\epsilon_i). \tag{2.108}$$

The difficulty comes from the fact that for a three-dimensional potential the level density increases with energy in a nonlinear way. This rise, however, does not go smoothly, but the levels are grouped in bunches roughly $\hbar\omega_0$ apart. If we imagine, for convenience of presentation, the level density smeared out with a Gaussian of width $\ll\hbar\omega_0$, then we get, schematically, the picture in Fig. 2.28. In which way, then, can the average part, and the oscillating part of such a density distribution, be separated?

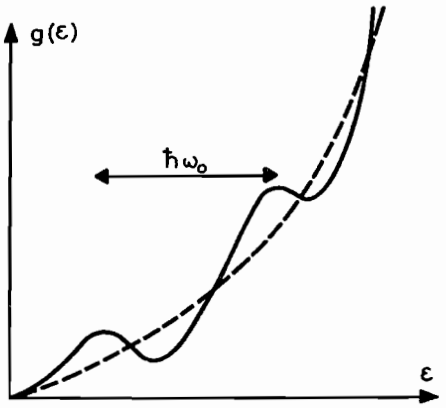


Figure 2.28. Schematic representation of the level density in an infinite three-dimensional potential.

As usual, we will think of the average density as given by a folding procedure

$$\tilde{g}(\epsilon)=\frac{1}{\gamma}\int_{-\infty}^{+\infty} g(\epsilon')f\left(\frac{\epsilon'-\epsilon}{\gamma}\right)d\epsilon', \tag{2.109}$$

and as a straightforward possiblity we could think of f as being a Gaussian with $\gamma\simeq\hbar\omega_0$. However, this is not sufficient, because an appropriate averaging procedure should leave the averaged level density $\tilde{g}(\epsilon)$ unchanged if averaged again with the same procedure, that is,

$$\tilde{g}(\epsilon)=\frac{1}{\gamma}\int_{-\infty}^{+\infty} \tilde{g}(\epsilon')f\left(\frac{\epsilon'-\epsilon}{\gamma}\right)d\epsilon'. \tag{2.110}$$

For f a Gaussian and g not a constant, condition (2.110) is clearly not fulfilled. In general, it will be difficult to do this for a given $\tilde{g}$ exactly, and it is not our intention to fulfill Eq. (2.110) for an arbitrary $\tilde{g}$, since then f would have to be a δ-function. Equation (2.110) will only be required to be fulfilled for smooth functions $\tilde{g}$. Since the essential contributions to the integral (2.110) come from the vicinity of the point $\epsilon'=\epsilon$, we suppose that $\tilde{g}(\epsilon)$ can be represented locally by a polynomial of degree $2M$ (usually $2M=2, 4,$ or 6). The condition (2.110) can then be fulfilled if one constructs f in the following way:

$$f(x)=P(x)w(x), \tag{2.111}$$

where $P(x)$ is an even polynomial of degree $2M$ and $w(x)$ is a weighting function like a Gaussian or a Lorentzian, for example.

We can check that Eq. (2.110) is satisfied in this approximation by [AS 65]

$$P(x)=\sum_{n=0}^{M} Q_{2n}(x)Q_{2n}(0), \tag{2.112}$$

where the set of orthogonal polynomials $Q_m(x)$ is determined by the condition

$$\int_{-\infty}^{\infty} w(x) Q_m(x) Q_{m'}(x)\, dx = \delta_{mm'}. \tag{2.113}$$

If we take for w a Gaussian

$$w(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}, \tag{2.114}$$

then $P(x)$ is given by a generalized Laguerre Polynomial [AS 65, Chap. 22]

$$P(x) = \sum_{n=0}^{M} \left(2^{2n} \cdot (2n)!\right)^{-1} \cdot H_{2n}(x) H_{2n}(0) = L_M^{1/2}(x^2), \tag{2.115}$$

where $H_{2n}(x)$ are the Hermite Polynomials. Thus we have explicitly constructed an averaging function f satisfying condition (2.110) for $\tilde{g}(\epsilon)$ which is locally approximated by a polynomial of degree $2M$. For practical applications, we give the coefficients of the polynomials $L_M^{1/2}(x^2)$ for $M = 0, 2, 4, 6$ (Table 2.5):

$$L_M^{1/2}(x^2) = \sum_{n=0}^{M} a_{2n} x^{2n}. \tag{2.116}$$

Table 2.5 Coefficients of the four lowest Laguerre Polynomials

M	a_0	a_2	a_4	a_6
0	1	—	—	—
1	3/2	−1	—	—
2	15/8	−5/2	1/2	—
3	35/16	−35/8	7/4	−1/6

There remains now the question of how to determine the precise values of γ and M. In general, the results will depend on these parameters. Since we want to take out of $g(\epsilon)$ the oscillation with the approximate frequency $\hbar\omega_0$, our method will only be meaningful if we can find a certain interval of reasonable γ-values ($\gamma \simeq 1$–$1.5\ \hbar\omega_0$) and corresponding M-values within which this dependence is practically negligible. We can say, then, that our results are independent of the averaging procedure (as should of course be the case). More precisely, this means that the averaged energy must show a "plateau" as a function of γ for fixed M within which the "plateau condition" [BP 73]

$$\frac{\partial \tilde{E}_{\mathrm{sh}}}{\partial \gamma} = 0 \tag{2.117}$$

is valid. Certainly for an arbitrary distribution of single-particle levels there will be no such plateau. But for the physically interesting distributions there always exists a certain bunching of levels with a frequency of roughly

$\hbar\omega_0$, so that we can expect that there is a plateau. (In practical calculations with finite potentials of the Woods–Saxon type, there are sometimes difficulties in finding a plateau. However, special methods have been developed to deal with such problems [BP 73, SI 75]).

In Chapter 13, we will show that there is a close connection between the Strutinski smoothing procedure and the semiclassical (Thomas–Fermi) method. This is not surprising, since it is known from the Thomas–Fermi theory of atoms that this method gives the relevant average quantities. We therefore urge the reader to consult Chapter 13 for a deeper understanding of this subject.

2.9.4 Strutinski's Shell Correction Energy

Having defined γ and the averaging function $f(x)$, we get the smooth level density from Eqs. (2.108) and (2.109).

$$\tilde{g}(\epsilon)=\sum_i \frac{1}{\gamma} f\left(\frac{\epsilon'-\epsilon_i}{\gamma}\right) \tag{2.118}$$

and the smooth part of the ground state energy in the shell model:

$$\tilde{E}_{\mathrm{sh}}=\int_{-\infty}^{\tilde{\lambda}} \tilde{g}(\epsilon)\epsilon\, d\epsilon. \tag{2.119}$$

The chemical potential $\tilde{\lambda}$ is determined by the condition (2.105)

$$A=\sum_i \tilde{n}_i=\sum_i \int_{-\infty}^{t_i} f(x)\, dx; \qquad t_i=\frac{\tilde{\lambda}-\epsilon_i}{\gamma}. \tag{2.120}$$

The $\tilde{n}_i$ can be considered as generalized occupation numbers. Therefore, for fixed A, Eq. (2.120) allows the determination of $\tilde{\lambda}$ through an iteration procedure. It is convenient to rearrange Eq. (2.119) a little:

$$\tilde{E}_{\mathrm{sh}}=\sum_i \epsilon_i\tilde{n}_i+F, \tag{2.121}$$

where F is then given by

$$F=\sum_i \int_{-\infty}^{\tilde{\lambda}} (\epsilon-\epsilon_i)\frac{1}{\gamma} f\left(\frac{\epsilon-\epsilon_i}{\gamma}\right) d\epsilon=\gamma\sum_i \int_{-\infty}^{t_i} xf(x)\, dx. \tag{2.122}$$

With this equation and Eq. (2.121), we get for the plateau condition (2.117)

$$\begin{aligned}\frac{\partial \tilde{E}_{\mathrm{sh}}}{\partial\gamma} &= \frac{\partial}{\partial\gamma}\sum_i \epsilon_i\tilde{n}_i+\frac{1}{\gamma}F+\gamma\sum_i \frac{\partial t_i}{\partial\gamma} t_i f(t_i)\\ &= \frac{1}{\gamma}F+\tilde{\lambda}\frac{\partial}{\partial\gamma}\sum_i \tilde{n}_i\\ &= \frac{1}{\gamma}F+\tilde{\lambda}\frac{\partial}{\partial\gamma}A=\frac{1}{\gamma}F.\end{aligned} \tag{2.123}$$

The last equality follows because the particle number A is a constant. Therefore, we find that the plateau condition (2.117) is equivalent to the vanishing of F.

Introducing the shell model occupation numbers

$$\begin{aligned} n_i &= 0 \quad \text{for} \quad \epsilon_i > \epsilon_F, \\ n_i &= 1 \quad \text{for} \quad \epsilon_i \leqslant \epsilon_F, \end{aligned} \tag{2.124}$$

we find for the total shell correction:

$$E_{\text{osc}} = \sum_i \epsilon_i (n_i - \tilde{n}_i) = \sum_i \epsilon_i \, \delta n_i . \tag{2.125}$$

Figure 2.29 shows the quantities δn_i for the actual case of a deformed Woods–Saxon potential and a Gaussian average with $M=2$ and a width of $\gamma = 6.6$ MeV.

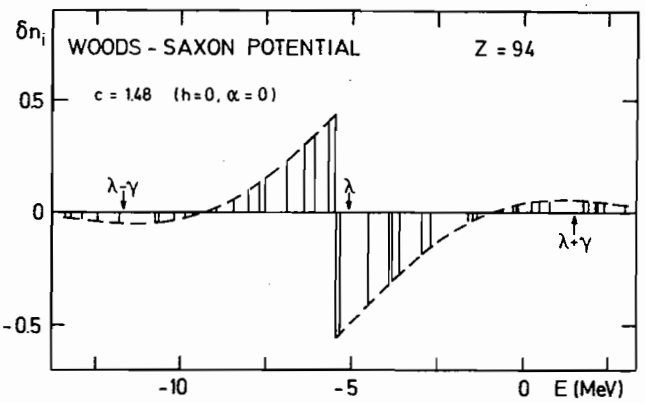


Figure 2.29. Deviations from the shell model occupation numbers resulting from a Strutinski averaging procedure for a Woods–Saxon potential. (From [BP 73].)

It is important to notice that the quantities $\tilde{n}_i$ are not actually occupation probabilities, as they can in fact have negative values. On the other hand, the $\tilde{n}_i$ behave like real occupation numbers and they indicate, for instance, how far we have to smear out the sharp Fermi surface of the shell model to get the smooth part of the ground state energy.

Having calculated the values δn_i, we are able to calculate the shell corrections to the density in configuration space [Di 71]:

$$\delta\rho_{\text{osc}}(\mathbf{r}, \mathbf{r}') = \sum_i^A \phi_i(\mathbf{r}) \phi_i^*(\mathbf{r}') \, \delta n_i . \tag{2.126}$$

This expression allows the calculation of averaged expectation values for all single-particle operators.

To give a definite example we will present the results of such a Strutinski calculation for the harmonic oscillator, which is naturally an over-idealized

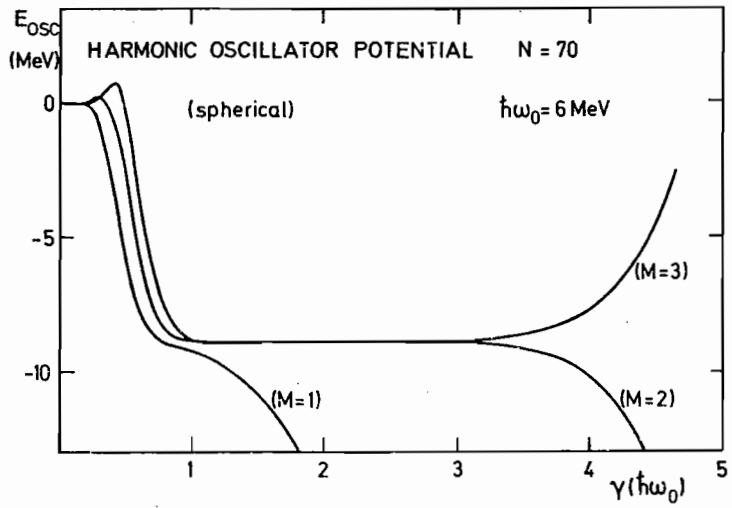


Figure 2.30. Shell correction energy for the harmonic oscillator as a function of γ and M. (From [BP 73].)

system, but which has the advantage of being rather instructive. In Fig. 2.30, the shell correction E_{osc} is shown calculated with a Gaussian weighting as a function of γ and M. We see that for $M \geqslant 2$ a well pronounced plateau develops at $E_{sh} = -8.91$ MeV.

It can also be shown that a plateau develops at precisely the same energy if we had taken $w = 1/2\cosh^2 x$ instead of a Gaussian for the weight function.

From the behavior of the values δn_i shown in Fig. 2.29 we conclude that only the shells in the neighborhood of the Fermi surface are important for the calculation of the shell correction. It is therefore usually sufficient to include three major shells in the averaging procedure. Only energies in the neighborhood of the Fermi surface are described well by the shell model, and it is exactly these which go into the shell corrected liquid drop energy. This is quite gratifying, since the phenomenological models are adjusted so as to reproduce the single-particle levels close to the Fermi level.

For the sake of completeness we have to mention that pairing correlations play an important role in heavy nuclei. They are usually treated in the BCS-model (s .e Chaps. 6 and 7). Since they are closely connected with the level density at the Fermi surface, one also observes oscillations in the pairing energy P. The liquid drop energy is adjusted to experimental masses. Therefore, it already contains the smooth part of the exact pairing energy, and we have to add only the oscillating part P_{osc}. It is obtained using the same philosphy as the difference between the pairing energy P_{BCS} in the BSC-model and its smooth part $\tilde{P}_{BCS}$. The total energy then has the form

$$E = E_{LDM} + E_{sh} - \tilde{E}_{sh} + P_{BCS} - \tilde{P}_{BCS}. \tag{2.127}$$

The smooth part of the pairing energy $\tilde{P}_{BCS}$ can be calculated with similar methods as presented here. For details the reader is referred to the literature [BDJ 72]. Another way to obtain the average properties for pairing correlations would be the extension of the Thomas–Fermi method to superfluid systems (see Sec. 13.2.6)

2.9.5 Shell Corrections and the Hartree–Fock Method

As we will explain in Chapter 5, there exists a theory which allows one to calculate the shell model potential microscopically. This theory is the Hartree–Fock method and its generalizations [HFB; see Chapter 7]. Modern calculations of this type give quite good and detailed agreement of the binding energies over the whole mass region, indicating that in those calculations the bulk properties are also well described. Nevertheless, one can still try to separate the smooth part and the oscillating part of the quantities calculated in this theory. The smooth part should correspond exactly to the liquid drop results, and it has been shown that this is in fact the case [BQ 75a, b]. As an example, Fig. 2.31 shows the deformation energy of the nucleus ^{168}Yb as a function of the quadrupole moment Q_2.

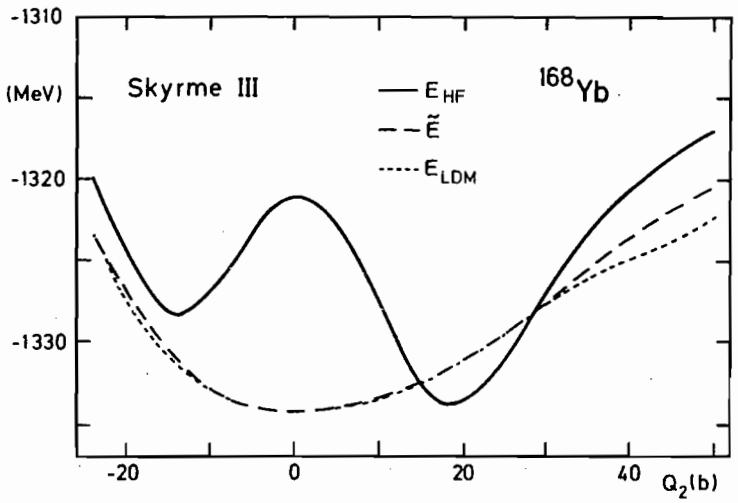


Figure 2.31. Deformation energies of the nucleus ^{168}Yb calculated with the force Skyrme III (see Chap. 4). E_{HF} is the exact HF-energy, $\tilde{E}$ the corresponding averaged part [Eq. (2.132)] and E_{LDM} is the liquid drop energy on the same path in the deformation coordinates Q_2 (quadrupole moment) and Q_4 (hexadecupole moment). (From [BQ 75b].)

To show that the Strutinski procedure based on a shell model potential gives, to a high degree of accuracy, the same results as a complicated Hartree–Fock

calculation, we now have only to prove that the oscillating part of the ground state energy in the Hartree–Fock theory is actually the same as the one derived in Eq. (2.125) from a shell model calculation. This we will do in the following, and we advise the reader who is not familiar with Hartree–Fock theory to return to the following somewhat subtle considerations after having progressed to the microscopic theories and, in particular, after having studied Chapter 5.

The Hartree–Fock ground state energy is, in a representation-independent way, given by* [see Eq. (5.28)]:

$$E_0^{\mathrm{HF}} = \mathrm{Tr}(t\rho) + \tfrac{1}{2}\mathrm{Tr}_1\,\mathrm{Tr}_1\,\rho\bar{v}\rho, \tag{2.128}$$

where ρ is the self-consistent single-particle density and $H = t + V$ is the total Hamiltonian of the system with t and V the kinetic and potential energies, respectively. The Hartree–Fock part h^{HF} of the Hamiltonian is a one-particle operator, and is defined by

$$h^{\mathrm{HF}} = t + \mathrm{Tr}_1\,\bar{v}\rho. \tag{2.129}$$

Now we can divide the density of ρ in the sense of Strutinski (Sec. 2.9.3) into a smooth and an oscillating part:

$$\rho = \tilde{\rho} + \delta\rho, \tag{2.130}$$

where $\delta\rho$ is defined by analogy with Eq. (2.126) for HF-single-particle energies.

Inserting this into Eq. (2.128) and collecting terms of different order in $\delta\rho$, we get

$$E_0^{\mathrm{HF}} = \tilde{E} + \mathrm{Tr}\,\tilde{h}^{\mathrm{HF}}\,\delta\rho + 0(\delta\rho^2), \tag{2.131}$$

where

$$\tilde{E} = \mathrm{Tr}(t\tilde{\rho}) + \tfrac{1}{2}\mathrm{Tr}_1\,\mathrm{Tr}_1\,\tilde{\rho}\bar{v}\tilde{\rho} \tag{2.132}$$

and

$$\tilde{h}^{\mathrm{HF}} = t + \mathrm{Tr}_1\,\bar{v}\tilde{\rho}. \tag{2.133}$$

Therefore, if the phenomenological shell model potential gives the same single-particle spectrum in the vicinity of the Fermi surface as the averaged Hartree–Fock potential, $\tilde{h}^{\mathrm{HF}}$, then the definitions of the oscillating part of the ground state energy, as given by the second term on the r.h.s. of Eq. (2.131) and by Eq. (2.125), agree if second-order terms in $\delta\rho$ can be neglected. These have been checked numerically and indeed turn out to be small corrections [BKS 72, BDJ 72, BQ 75a].

If the phenomenological single-particle spectrum does not give exactly the same single-particle spectrum as the operator $\tilde{h}^{\mathrm{HF}}$, the above statement nevertheless stays true, because small deviations again give contributions of second order in Eq. (2.131). Equation (2.131) is also called the *Strutinski energy theorem* [St 68, St 74]. It states that all shell effects of first order in $\delta\rho$ are taken into account by summing up single-particle energies of an averaged single-particle Hamiltonian.

In this sense, the Strutinski procedure provides a method which reproduces microscopic results in an optimal way using phenomenological models. Needless to say, the latter are much easier to handle for realistic calculations.

* The definition of Tr_1 is given in Eq. (E.19).

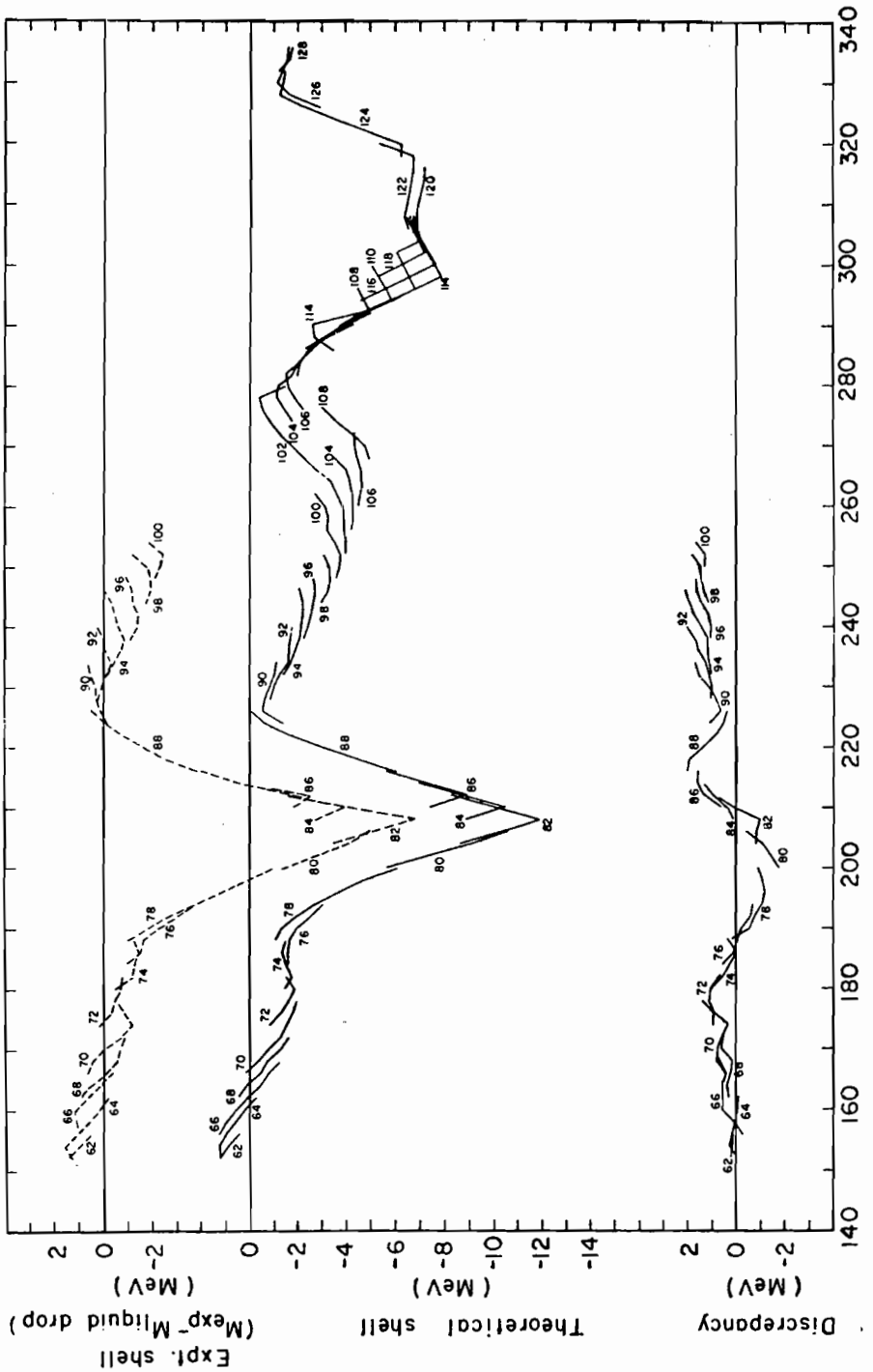


Figure 2.32. Comparison of theoretical mass corrections with experimental values. (From [NTT 69].)

2.9.6 Some Applications

The combination of the shell model with the liquid drop model as provided by Strutinski's theory has, for instance, been applied to the calculation of nuclear masses [NTT 69]. Figure 2.32 shows the theoretical mass corrections to the liquid drop model and compares them with the experimental data. The overall agreement is very good; the oscillating structure in the vicinity of the double magic nucleus ^{208}Pb is especially well reproduced. However, we must realize that these shell corrections were calculated with Nilsson's single-particle energies, which do not reproduce the experimental single-particle level scheme of ^{208}Pb (see Fig. 2.15) very well. In fact, the shell correction calculated with these experimental levels is 5 MeV off in ^{208}Pb. For a discussion of this discrepancy, see [BDJ 72].

Another application of the Strutinski method is the calculation of energy versus deformation curves. As we mentioned briefly at the end of Section 2.8, the calculation of the energy as a function of not only the quadrupole deformations but also octupole, hexadecupole, and higher deformations gives rise to multidimensional energy surfaces. As in the shell model, we find deformed minima in the rare earth nuclei as well as in the actinides, and sometimes second minima for very large deformations (fission isomers). It turns out that the ground state minima are reproduced roughly at the same deformations as in the pure Nilsson model. However, at large deformations, the deformation surfaces look different (for a detailed investigation, see [BDJ 72]). In these surfaces one looks for the path which goes from the equilibrium deformation past the lowest barrier and out to the region of deformation in which fission occurs. The knowledge of this path and its actual shape for a nucleus then allows an estimate of its fission lifetimes [ADD 70, JNS 70, BFN 72, Ni 72, BDJ 72, Pa 73, LP 73, MN 73, RNS 78]. In particular the stabilities of superheavy nuclei have also been investigated in this way (see [Ni 72, NN 74, BN 77] and references given there).